

WHY DO WE DIE?

T. BODLEY SCOTT

T. FISHER UNWIN LTD
LONDON

WHY DO WE DIE?

T. BODLEY SCOTT

T. FISHER UNWIN LTD
LONDON

The Project Gutenberg eBook of Why do we die?

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Why do we die?

Author: T. Bodley Scott

Release date: April 28, 2023 [eBook #70660]

Language: English

Original publication: United Kingdom: T. Fisher Unwin Ltd., 1921

Other information and formats: www.gutenberg.org/ebooks/70660

Credits: Thiers Halliwell, Susan Skinner and the Online Distributed Proofreading Team at <https://www.pgdp.net> (This book was produced from images made available by the HathiTrust Digital Library.)

*** START OF THE PROJECT GUTENBERG EBOOK WHY DO WE
DIE? ***

Transcriber's notes:

The text of this e-book has been preserved as in the original, but the varied layout of numerous advertisements at the end of the book has been standardised.

WHY DO WE DIE?

THE ROAD TO A HEALTHY OLD AGE

BY T. BODLEY SCOTT, M.R.C.S.
(Eng.).

Author of "The Religion of a Doctor."

SECOND EDITION.

Crown 8vo, cloth, 5/- net.

Third Impression.

"In this book an attempt is made to demonstrate both to the medical profession and the laity that premature decay, physical and mental, may within limits be prevented. . . . We have perused the book with pleasure, and cordially recommend it to our readers."—*Medical Times*.

"We can cordially commend this work to readers of advancing years, whether medical or lay. To the former it presents an old subject in a new way, and to the latter will convey some very wholesome material for thought."—*Lancet*.

T. Fisher Unwin Ltd. London.

WHY DO WE DIE?

BY

T. BODLEY SCOTT

M.R.C.S. ENG., L.R.C.P. ED.

T. FISHER UNWIN LTD.
LONDON: ADELPHI TERRACE

First published in 1921

(All rights reserved)

PREFACE

The attitude and the pretensions of the “medicine man” may perhaps have had some value in the days when the multitude was full of superstition and had no scientific knowledge, but we hope those days of ignorance have passed, though we fear superstition remains. The “medicine man” was a more or less clever poseur, who used his small amount of knowledge to conceal his very large amount of ignorance. The medical profession, till the more recent years, cannot be entirely acquitted of the same sort of dishonesty, but a new era of open-mindedness and honesty has, I think, begun.

To get any real success in the treatment of disease, we must get confidences, not one-sided, but on both sides, between ourselves and our patients, and so alone shall we get that co-operation without which there can be little expectation of success. At present we see this success more in the wonderful results of psychotherapy than in the treatment of objective disease, but I firmly maintain that mutual trust will prove its great value in this also. Is it reasonable to expect that men of equal learning and intelligence as ourselves will accept our treatment and advice without knowing the rationale of them? Should we do it in the case of legal or religious matters? Our patients know as well as we do that medical knowledge is in a continuous state of flux and that as yet we are nowhere near finality. The dazzling garment of omniscience will be discovered to be the somewhat shop-worn cloak of ignorance. A wise and observant patient can help us enormously in the investigation of a difficult problem in disease, for the subjective symptoms are often as important as the objective. What the eye does not see may be there all the time.

The history of the world right up to the present time has shown the futility of dividuality as against individuality both in politics and religion. The greater part of our Christian teaching is against separateness, and towards

mutual love, consideration, and help. For this reason I have tried to write so that all may grasp the hope and possibility of a longer and more effectual life. Co-operation is the watchword of the future, isolation the disease and explanation of the past.

GLOSSARY OF MEDICAL TERMS

There are parts of this book that can only be fully grasped by a physician, but I have tried to write it so that any intelligent reader may be able to see not only the drift of the argument, but may be able to make a personal application of it. To this end I am giving a glossary with definitions of the more abstruse scientific words.

Pathological.—The branch of medical science that treats of diseased conditions, their nature and causes.

Ventricle.—The left ventricle is the chief pump of the heart which by its powerful contraction forces the blood forward into the aorta, the first part of the arterial system. It is the initial motor of the circulation.

Systole.—The contraction of the heart that drives the blood onward.

Diastole.—The dilating time of the ventricle in which the blood flows in to fill it for its next contraction.

Sphygmomanometer.—The name of the instrument that measures the blood pressure.

Peripheral.—The parts farthest from the centre.

Tinnitus.—A singing or ringing in the ears.

Hyperpiesis.—Abnormal increase of blood pressure.

Atheroma.—A form of fatty degeneration with hardening of the inner coats of the arteries.

Endocrine.—A name given to certain glands of the body which have no external outlet or secretion (ἐνδον, within, and κρίνω, to separate), such as the thyroid.

Protein.—The entire amount of nitrogenous material in organic substances.

Hypophysis.—The pituitary gland situated at the base of the brain.

CONTENTS

	PAGE
PREFACE	<u>5</u>
GLOSSARY OF MEDICAL TERMS	<u>9</u>
CHAPTER I	
THE CHOICE OF WAYS	<u>15</u>
CHAPTER II	
ARTERIO-SCLEROSIS: ITS NATURE AND ITS CAUSES	<u>27</u>
CHAPTER III	
THE WAY OF SAFETY	<u>65</u>
CHAPTER IV	
THE MEDICINAL TREATMENT OF SCLEROSIS	<u>101</u>

CHAPTER I

THE CHOICE OF WAYS

Why do we die so long before our time? And what do we die of? These are the two most crucial questions, as far as this life is concerned, that affects us all, but which most of us seem content to ignore.

Putting on one side the accidents of life—and among the accidents I include the acute diseases, such as pneumonia, rheumatic fever, and the infectious diseases, which attack from without, and the so far unexplained cancer—we must allow that our lives and our health are to a great extent in our own keeping, and of our own making. Three score years and ten was David's estimate of the working years of a man's life, but under modern conditions of improved sanitation the term of life should be at least four score years and ten, and this extra 20 per cent. should not be a time of pain and sorrow, but of peace and rest, with still some capacity for work.

Many of us used to be content to answer the first question, "Why do we die?" by saying our time had come. The unreflecting but so-called religious person seemed to think that God had placed a time limit against each individual when he came into the world. This is pure fatalism, and unworthy of the name of reason or of faith in the Divine justice. "Let us eat and drink, for to-morrow we die" is almost the logical position of such minds. We cannot hope to solve individually or in detail the mysteries of life and death, why one is cut off in the full tide of his energy, and another, who seems to be going out on the ebb, remaineth. But we must put these following questions honestly to ourselves—Are free will, scientific work, knowledge, self-restraint, and self-denial to be so much wasted powder? Are we to work out our destiny, the sport of chance, hopelessly, illogically, without ambition and without a healthy pride in our work? Is the man who spends his life in riotous living, in over-eating and drinking, and in

committing every physiological sin that he can think of, to live as long and as healthily as the wise and sober man? Has the great Father of us all given us no motive for life, no work to do for Him or for our brothers?

Must not we, as separate sections of the Divine image and of His all-pervading spiritual force, endeavour with all our strength and knowledge to prolong the life given us to the utmost limit? Surely the fact of our life, the gift of our life, becomes a great personal responsibility, an acknowledged debt to our Maker that we can only repay by regarding it as a treasure to be daily cared for and cherished. Thus the promotion of health and the prevention of disease must manifestly be our first considerations, for without health our output of work ceases, or becomes both in quantity and character a poor thing.

The despairing cry of “Why was I born?” comes to most of us in the sorrows and tragedies of life, but it is the cry of faithlessness, of relativity, of blindness to the sure, but half-seen purposes of the Absolute God.

That which we in our presumption are pleased to call civilization, should certainly further such aims for health, but the whole story of the world, of past and gone kingdoms, and of our own times also, show that civilization works up to a point of luxury that defeats and destroys itself. Babylon, Egypt, and Rome all testify to this end. These old civilizations perished, perhaps, more from enervating luxury and dissipation than from violence; though some of them gave way through feebleness, physical and mental, to a stronger, hardier race from without. In our own times there seems to loom the same danger, but the attack comes from another side. We neither eat nor drink to such excess as our forefathers, and we are not grossly luxurious or debauched, but the strain and the competition of life, both in professions and in business, has become so intense that our nervous systems break down long before their time. This produces rather a new type of an old disease. What is this disease but arterio-sclerosis?—a word which means only thickening of the arteries, but which, after fifty, in its far-reaching results destroys life and working capacity far more than cancer or any other disease. And this disease is, in the great majority of people, the result of the

almost incessant nervous strain that seldom allows our hearts and our arteries, and especially the arteries of our brains, to get their proper quota of rest.

Though the average duration of life has increased during the last fifty years, much of this is due to a lower mortality in infancy and childhood. There has been an actual increase in the mortality rate among males between the ages of fifty-five and sixty-five. One-third of the total deaths in this period are due to diseases of the heart and blood vessels. In America it is stated to be 10 per cent. higher. If the causes of death in certificates were given more exactly, this proportion would be without doubt still greater. These deaths cannot be fairly brought under the heading "Old age." This is what we die of! According to our death certificates we die of paralysis, of strokes, of Bright's disease, of softening of the brain, of cardiac degeneration, of angina pectoris, but in nearly every case the foundation, the *causa causans* is arterial disease.

Sir Clifford Allbutt, in his distinctive way, says:—

"We recognize now, all of us, that, in the lapse of men's years, one long reckoning of his mortality is, and from all known ages has been, written on the walls of his vessels. We may suppose that in the primitive man, by external conditions if not by innate capacity, life was of comparatively brief duration. Domestic animals seem, as a rule, not to live long enough to use up their arteries or not to live ill enough to abuse them, and amongst these creatures atheroma, though not unknown, has not been commonly observed. We may reasonably conjecture that, as in Stable Societies human life became protracted, the arteries got time to wear out."

There is plenty of evidence from the mummies of Egypt that atheroma and sclerosis were common, not only in advanced age, but in the middle years of life. We cannot know intimately the life that these old people lived, but we can pretty well picture from history both the height of luxury and the nervous strain. Civilization brings without doubt more comfort, more luxury, and more scientific knowledge, but it seldom brings more true

personal wisdom. The philosophical calm and contentment with simple pleasures, and wise simplicity of life, are still the exception. To use a slang expression, we are nearly all of us “on the make,” stretching forth eager hands and hearts towards the will-o’-the-wisp of wealth and advancement, while we let the real values and beauties of life pass us by and so we continually miss our markets. The great war has taught us one good lesson, viz., that we can live and be well—nay, better—on much less food and on simpler than was our former custom; but it has increased, at any rate for the present, the struggle for and the pain of mere existence.

We must nevertheless be optimistic, and hope that the great advance in science may soon make life easier, and the food of the world once again sufficient for its inhabitants; then, please God, materialism may shrink into the background and things spiritual come into their own. Not, however, by a reformed and wiser life alone do I hope to show the more excellent way, but to show that modern science has placed in our hands a preventer of decay and a prolonger of life and energy. I cannot promise that this discovery will take twenty years off a woman’s life and put twenty years on to her beauty, but I know that it will give many more years of fruitful and enjoyable work, with greater strength and better health.

CHAPTER II

ARTERIO-SCLEROSIS: ITS NATURE AND ITS CAUSES

The thickening chiefly of the middle coat of the arteries that we find in this disease is the first pathological sign, and it implies that morbid structural changes have begun; but long before this occurs, the predisposing causes have been at work. The actual structural change is the visible effect of the law that over-use of a muscle produces over-growth or hypertrophy. This is not a law of disease, but of purposeful utility, designed to meet emergencies of life. Still there is a further law in nature, that long-continued excessive use of any muscle leads to a form of hypertrophy which cannot maintain its vitality, but degenerates. Arteries must be treated as muscles, for their muscular coats contract and dilate, as circumstances demand, under the control of the nervous system.

Dr. George Oliver, in his third edition of *Studies in Blood Pressure*, among his fundamental data lays down this law:

“Though the left ventricle—dispensing its blood under a high pressure into the aorta—is the master hand in maintaining the arterial blood pressure, the height to which that pressure rises depends primarily on the resistance encountered in the peripheral arterial system, and in the capillaries, and, secondly, on the response of the ventricle to overcome that resistance. Given a normal ventricle the arterial tonus largely determines the ventricular response and the level of the arterial blood pressure; and this is the keynote of the physiological and pathological variations of that pressure.”

Arterial tension may be described thus: During the resting-time of the heart, between each beat, the circulation of the blood is maintained by the contractile power of the arteries, which steadily urges the blood onwards

into the capillary vessels, and so into the veins. This power is known as arterial tension or pressure, and it is estimated fairly accurately by the sphygmomanometer. A large number of observations have taught us what is the average pressure at different ages of life, and thus we are able to talk about a normal or abnormal pressure.

If we hope to understand and successfully treat abnormal pressure and sclerosis, we must visualize the whole of this picture clearly, wisely, and in its due proportions. This we have not always done. The heart man, as Osler wittily said, “tinkers at the pump,” while the pressure man, with his manometer, concentrates on the pipes. To give a scientific description and explanation of the physics of the circulation is far beyond me, and would be inappropriate in a small practical work of this sort, but a study of some elaborate standard work, such as Sir Clifford Allbutt’s *Diseases of the Arteries*, or Halliburton’s *Physiology*, would do much to give a fair understanding of this intricate subject: but, as they so often acknowledge, the vital problem of the circulation, with its ever-changing demands, both general and local, can hardly be solved by the known laws of physics. Heat and cold, joy and grief, anger and fear, acting through the nerves of the vaso-motor system, all help to complicate the problem and often to confuse cause and effect.

As far as we can as yet judge, the first objective symptom that points to threatening sclerosis is a more or less continuous rise of systolic pressure, as measured by the sphygmomanometer (the use of this instrument is imperative). The following extract from the *Lancet* January 8, 1916, puts the matter very clearly: “The clinical value of observations on Blood Pressure is doubted only by those who have never made them. It is true that the finger can detect some difference in pulse tension, but it is often entirely at fault, since it can only estimate total pressure, thus missing a high pressure, if the volume be small. In the same way, although the hand can detect differences of temperature, it cannot replace the thermometer. The sphygmomanometer has enlarged our ideas, cleared up difficulties in diagnosis, and helped in prognosis. This has naturally reacted on treatment;

it has enlarged our ideas on 'heart failure.' The heart is adjusted to work at certain pressures, and though within considerable limits it can adjust itself to variations, those limits may be exceeded."

There are subjective signs, especially in middle age, which point the same way, but we can seldom get hold of patients at this early stage. These are chiefly shortness of breath, on slight exertion, which makes a man realize that he has almost suddenly got out of condition, that the elasticity of youth has gone; a feeling of brain fatigue, quite out of proportion to the work done; sometimes giddiness and tinnitus, especially on stooping. Whether these symptoms appear before there is a rise of pressure it is difficult to say, but they should certainly in all cases suggest blood pressure testing. The key to the patient's future then lies first in our hands, and then in his, life or death, health or disease, a full working life and happiness or incapacity and failure. The examination must be carefully made and frequently. The best time to take pressure is about two hours after food, when there has been a time of physical quiet, and no hurry or emotion. The position should be either recumbent or sitting in a chair, with the arm resting on a table (the muscles of the arm should not be in action), and the wrist should be on the same level as the heart. Pressure readings are higher when standing than sitting, and when sitting than lying down, the difference being from four to eight millimetres. In the early stages pressure varies a good deal, and will not show a rise at all times of day; in the early morning particularly it is often quite normal.

The question now arises, What is the normal? For our purpose we may leave out the question of pressures in childhood and early life, and may take the following figures as a fair average for our calculation:—

From 40 years of age to 50, 130 mm. to 135 mm.

50 " " 60, 135 mm. to 145 mm.

60 " " 70, 140 mm. to 150 mm.

I think there can be no doubt that there is a period, often of months, when there is only increased contraction of the muscular coats of the arteries. This

is due to the increased strain of life generally, and is in consonance with our experience generally of muscular overgrowth and overwork. Huchard called this "the pre-sclerotic stage."

I think we may say that any pressures at about fifty years of age below 120 mm. are a sign of debility, and above 150 mm. are signs of threatening arterial trouble. Over sixty there is a natural tendency to hardening of the arteries, and many men seem to lead healthy lives with a pressure of 160 mm. Still, with a pressure of that height they cannot afford to put any excessive strain on their arteries. At about eighty there is a decided tendency for pressure to fall, even if there has been plus pressure before. It is rare to find high pressure over ninety for the reason that people who live to that age must have had healthy hearts and arteries all the time, but even in them we not infrequently find passing rises; these are often missed, but they cause much mental distress and even delusions, but at this age, under treatment, the pressure soon falls and the mind clears. I had one lady who lived to ninety-seven, and during the last six years of her life she had visions and slight delusions when her pressure rose, but in the intervals her mind was quite clear. She learned to recognize the cause of her trouble.

This rule of normality stands pretty true for the majority of people, but in practice we find a good many individual exceptions: in such, the personal equation has to be solved. Women often, even in advanced age, have very low pressures, even 120 mm. to 125 mm., and seem to carry on quite well. They again get a period of high pressure about the climacteric, which with careful treatment disappears, and their subsequent years from about fifty-five to seventy-five are often the best and the happiest working years of their lives. It is not uncommon both in men and women who have led strenuous but healthy lives to find, about sixty, steady pressures of 160 mm. to 170 mm., without causing any symptoms of illness or distress. These people, though one cannot consider them safe lives from the insurance point of view, have learned to manage themselves and seem to have set up a new, personal normal which is an effective compromise. If one has been able by careful observation to satisfy oneself about such a case, it would certainly

be unwise to try to lower that pressure point. Even in cases where there is manifest disease of some standing, one has to find out the best working point of pressure for the individual. They can live and work fairly comfortably at 160 mm. or 170 mm., but if you lower the pressure to 140 mm. or 150 mm., their hearts get irregular and they feel good for nothing; but I think we may make a working rule that a pressure of 180 mm. and over should be lowered; the danger to the cerebral arteries is too great. Even in the early stages a careful watch should be kept on the kidneys. If albumen appears, it makes the prognosis more grave, but it should cause no despair. Mahomed and the early observers thought that kidney disease was almost inevitable sooner or later, but further experience has shown us that many cases live for years and then die without albuminuria. In other cases it is often transient and responds to treatment.

Hitherto I have only dealt with systolic pressure, but the diastolic pressure in senility and sclerosis is a matter of much importance, and helps us both in treatment and prognosis. Brunton said: "The systolic pressure shows the maximum height to which blood pressure is raised by the wave of blood driven into the aorta by the contraction of the left ventricle. It thus indicates in a general way the strength of the ventricle. The diastolic pressure shows the minimum to which the blood pressure sinks during the interval when no blood is coming into the aorta from the heart, and the arterial system is emptying itself through the capillaries into the veins. It therefore indicates generally the degree of contraction or relaxation of the capillaries."

For practical purposes we want to know the pulse pressure, and this is roughly estimated by subtracting the diastolic from the systolic. A systolic pressure of 140 and a diastolic of 100 would give a pulse pressure of 40, which is about right. Any pulse pressure of over 40 points to hyper-tension, and under 20 to hypo-tension. The diastolic pressure is more constant than systolic pressure, and is not so susceptible to disturbing external influences; it is therefore a surer guide in estimating hyper-tension. A continuous diastolic pressure of over 100 certainly points to continuous hyper-tension. If the difference between the systolic and the diastolic is normal, about 30

to 40, we may conclude that the heart's impulse is normal, and that even in conditions of marked sclerosis there is efficient compensation.

Sir Clifford Allbutt says (in *Diseases of the Arteries*, vol. i, p. 92): "It is in the experience of us all that, as the walls of the great vessels lose their elasticity, the systolic pressure mounts above its proportion to the diastolic which falls. In aortic regurgitation the difference between the systolic and diastolic figures may be enormous, as much as 105 mm. to 210 mm. In a certain 'Adams Stokes' case of my own, at a pulse rate of 32, the systolic pressure was 180, the diastolic 90 mm. Thus, with the difference between the two pressures waxing as the resilience wanes, the blood pressure moves more suddenly and largely, bangs more, and thus racks the machinery. The greater vaso-motor lability in youth has a like effect, but the vessels are then much more resilient and energy is stored. In an early hyperpietic, or nephritic subject the systolic exorbitancy may be considerable or variable, while the diastolic is steadily in excess. As age advances the amplitude increases to 80 and upwards; a systolic pressure of 160 mm. may be associated with a diastolic of 75 mm. or 80 mm."

Dr. H. C. Mann says: "That as compensation fails, the diastolic pressure begins to fall, and the systolic becomes irregular." It must seem very difficult for a beginner, or even for an experienced busy practitioner, to find his way through all these devious by-paths, which must look to him often to be blind alleys and to lead nowhere, but out of the welter a few good and sound working principles emerge.

"*Firstly*, that at middle age and afterwards the knowledge of an habitual systolic pressure is of first-rate importance.

"*Secondly*, that this knowledge can be confirmed and sometimes rectified by knowledge of the diastolic pressure (manifest valvular diseases of the heart, especially aortic regurgitation, are excluded); and that a continuous rise of systolic and diastolic pressure point conclusively to threatening or confirmed arterio-sclerosis.

“*Thirdly*, that a steady ratio between the two pressures, even when much above normal, is a sign that the heart is doing its work well; and conversely, a falling diastolic ratio is a sign of failing compensation.”

Dr. Louis M. Warfield, in his work on Arterio-Sclerosis, says: “It is most important to estimate accurately the diastolic pressure as well as the systolic, for only in this way can we obtain any data of value regarding the driving power of the heart and the condition of the vaso-motor. A high systolic pressure does not necessarily mean that a great deal of blood is forced into the capillaries. Actually it may mean that very little blood enters the periphery. The heart wastes its strength in dilating constricted vessels without carrying on the circulation adequately. The pulse pressure is the difference between the systolic and diastolic pressure, and practically represents the heart lode.”

All these considerations demonstrate the importance of watching closely the pump as well as the pipes.

In the above figures I am presuming that the auscultatory method of determining pressure has been used. It must also be manifest that, the earlier treatment begins, the more sure will be the success. It may be that all our theories will meet the ultimate fate of so many others and be abandoned, or partially disproved, but the practical therapeutic results will remain—an increased capacity for work and considerable prolongation of the later years of life.

For a short and graphic description of the two methods of estimating pressures I am poaching from an excellent book called *Blood-pressure Technique Simplified*, by Dr. W. H. Cowing, of America (pp. 12–15):—

“There are two methods of determining blood pressure, that by oscillation, and that by auscultation.

“We will first describe the method by oscillation.

“Place the bag over the arm with the two tubes well under the arm and over the brachial artery. Wrap the remainder of the sleeve around the arm much

the same as you would apply a bandage, tucking at least six inches of the sleeve under the last fold. Then place the sphygmomanometer in one tube and the bulb in another, and you are ready for reading. Care should be taken not to put the sleeve on tight enough to cause any apprehensive feeling in the patient.

“Place the finger lightly over the radial artery and send the pressure in the cuff to the point where the pulse has entirely disappeared, then advance the hand about 2 cm. above this point of disappearance, then release the air gradually and note the first perceptible pulse wave felt by the palpating finger at the wrist. This represents the true systolic or maximal pressure.

“Having now obtained the systolic or maximal pressure, release the air gradually by means of the escapement valve, and note where the largest oscillation or movement of the hand takes place, and the lowest point of this largest oscillation is the diastolic or minimal pressure.

“For example, if the largest oscillation occurred between the divisions 90 to 96, 90, being the lowest, would be the reading.

“It is well to bear in mind to keep the eye concentrated on the scale divisions, for the travel of the hand can be accurately determined when this is done.

“In determining systolic (maximal) pressure, the pulse becomes more feeble as the pressure advances, and when taken with the ends of the fingers, the pressure of the fingers is involuntarily increased, so that a very sensitive pulse may be closed off entirely by the finger pressure; while, if taken with the ends of the fingers resting on the upper curve of the bone of the wrist, permitting the balls of the fingers to rest lightly over the radial, thus bringing in contact with the pulse the most sensitive part of the fingers, any extra pressure that might result would be directed against the bone and not the pulse.

“*Method by Auscultation.*—This is by far the most accurate method of determining the blood-pressure, and the results obtained in this way should

be free from error, or the dangers of personal equation in palpating the artery at the wrist.

“Systolic or Maximal Pressure.—Bare the arm and adjust the sleeve well up; place the stethoscope over the brachial artery about 1 cm. below the border of the cuff, then constrict the arm by inflating the bag until no sound is heard in the artery through the stethoscope. At this point release the air slowly by means of the escapement valve, and soon a clear, clicking sound will be heard, which indicates the first passage of the blood stream below the constricting arm band and is the true systolic or maximal pressure.

“Diastolic or Minimal Pressure.—Still gradually releasing the air, this clear, clicking tone (first phase) is followed by a low tone or murmur (second phase), and this murmur is followed by a loud, clear tone (third phase), this in turn by a perceptibly dull tone (fourth phase), and this dull tone represents the diastolic or minimal pressure.

“Below are given the five different phases of tone to be heard in the auscultatory blood-pressure phenomena as found by Louis M. Warfield, A.B., M.D., and confirmed by his clinical observations on animals.

“The first phase is due to the sudden expansion of the collapsed portion of the artery below the cuff and to the rapidity of the blood flow. This causes the first sharp clicking sound which measures the systolic pressure.

“The second, or murmur and sound phase is due to the whorls in the blood stream, as the pressure is further released and the part of the artery below the cuff begins to fill with blood.

“The third tone phase is due to the greater expansion of the artery and to the lowered velocity in the artery. A loud tone may be produced by a stiff artery and a slow stream or by an elastic artery and a rapid stream. This tone is clear cut, and in general is louder than the first phase.

“The fourth phase is a transition from the third, and becomes duller in sound as the artery approaches the normal size.

“The fifth phase, no-sound phase, occurs when the pressure in the cuff exerts no compression on the artery and the vessel is full throughout its length.”

He also says that blood pressure always depends on four factors, viz.:

1. Cardiac energy.
2. Peripheral resistance.
3. Elasticity of the arterial walls.
4. The amount of blood in circulation.

It is very important to remember these and to give due value to each of them. With regard to No. 4, it is doubtful if this is of as much importance as has been thought or as one would at first think; the mass of blood in the body has such a large reserve of stowage room in the muscles and in the internal viscera, that its effect on general arterial pressure must be slight. The viscosity of this fluid is probably of far more importance. If one bleeds a case of high pressure freely, the pressure will fall, but only for a short time; the blood is thinned, but the loss in mass is quickly restored by absorption from the watery tissues of the body, and in a very few days the viscosity returns to its former point. Bleeding is thus very good for emergencies, such as threatening apoplexy or cardiac dyspnoea, but is of little permanent use in the routine treatment of high pressure.

Sir Clifford Allbutt, wisely and correctly, I think, divides most examples of abnormal high pressure into hyperpiesis, which describes this symptom alone, and into senile atheroma. Clinically this is important, for hyperpiesis, which generally comes on about fifty to sixty, is a very serious disease, which, untreated, destroys life in a few years, whereas senile atheroma may go on till advanced old age, crippling energy perhaps, but in many cases hardly shortening life at all. The former generally ends in cerebral hæmorrhage, cardiac degeneration or Bright's disease, but senile atheroma may escape all of these. In hyperpiesis raised pressure is continuous and tends steadily to increase; in atheroma there are times of high pressure, but

as the disease advances the pressure is often normal or subnormal, its gravity depending much on the seat of the atheroma.

Are we justified in regarding arterio-sclerosis as a true disease? I think not. One cannot justly compare it with entity diseases, such as the acute infections, as cancer, as tuberculosis, or the diseases that affect the brain and the spinal cord. It is, I think, one of the penalties of an artificial over-civilization, of social, business and competitive pressure, unwise in quantity and unscientific in method. Unfitted, as yet, in development, and lacking the necessary knowledge, men strive after a higher stage of evolution, or after illusions which they think to be so. For a while they seem to succeed, but involution and premature degeneration are the inevitable consequences. To attain their "ambitious" ends they live and work intensely, stoking their furnaces with far too much food, alcohol and other unnatural stimulants. Intemperate in work, intemperate in living, prodigal in waste, they come to an untimely end.

Such is the description of one type of case that we often meet, but there is another and a sadder class, where unfavourable circumstances through the greater part of life, pressure of poverty and ceaseless anxiety for others, produce a careworn existence, with no joy or colour in it, and often insomnia. Such as these, in spite of a sober, abstemious life, often get sclerosis. Their pulses are small and thready, but hard, and often deceive the finger. Paralysis from small cerebral hæmorrhage is the frequent end. We may take it as an axiom that happiness lowers pressure, while grief and care raise it. We must all of us feel this as a personal truth, though we may not be able to test it with a manometer. It is not only the cat that care kills. In melancholia the pressure is almost always well above normal; in maniacal conditions generally below.

Few of us know how to work with our brains; psychology has as yet taught us but little of method and of practical value, though for this we must blame ourselves rather than the psychologists. We drive our conscious minds with few or no intervals of rest till brain exhaustion comes; then the machinery gets out of control and the screw races; so come insomnia and poor,

confused results, with disappointment and depression. We talk glibly about the sub-conscious mind, and are now and again struck with some evidence of its working existence, but we few of us use it systematically as our servant—the most competent servant that exists, and one that never gives notice as long as life lasts. Our upper story, in which we suppose our conscious-thought machine works, is like a loft full of noisy machinery that often needs oiling and that often gets out of order. Its physical motor power gets slack or runs right down, and the output is poor or ceases. Sleep and rest alone will restore it and set it going again.

Our sub-conscious mind seems to be almost independent of our physical life; it works more or less continuously while the conscious mind is working, fooling or sleeping. To continue the loft simile: beneath the machine loft, with its clatter and dust, is a quiet darkened room, in which our sub-conscious mind lives and functions in its own quiet way. Men who know how to use it say that, when their conscious minds become exhausted and have come to an impasse, all they have to do is to open the trapdoor and pass the apparently insoluble problem down into the quiet room below. They then banish the subject from their minds completely, rest, play, or take up some other line of thought, and by and by, in a day or two perhaps, when they are not thinking of it, the problem turns up solved. There is no sense of fatigue or of conscious effort in this process: it is just the old adage of sleeping over a difficulty. Many of the world's great inventors have, knowingly or sometimes unknowingly, made their great discoveries in this way. Our conscious minds are limited in motive power and always subject to fatigue, but our sub-conscious minds rarely tire and are unlimited in scope and power, for they are a part of our Divine inheritance, a portion of the great undivided spiritual force that moves and governs the universe, that slumbers not nor sleeps. To get into close touch with this power needs persevering practice, but it is clearly of inestimable value. We must not expect a clear and useful response unless we have given concentrated attention to the matter in point; a mass of ill-digested, uncertain data will remain as such; we must be perfectly honest with our *alter ego*; it is of no

use trying to take him in. The more often and thoroughly we accustom what we call ourselves and our sub-conscious minds to work together in a friendly way, the better and the quicker will be the response.

The above is only a cursory sketch of a profound subject, but will serve to show what great economies can be made in the working of our limited human understandings.

The condition of our arteries that we call sclerosis is much more common in the sedentary brain-worker than it is in the outdoor man. We see it rarely in the farmer, the sailor, the labourer, unless it is as the result of habitual intemperance; we see it rarely in the athlete. The average blood-pressure of the indoor worker is certainly higher than that of the open-air worker, even in perfect health.

It follows, therefore, that the more we systematize and economize in our brain work, the less the strain on the arteries that supply that organ with its life-blood. Few of us realize what can be done in this direction. A much advertised system, which I need not further particularize, has got hold of some of the right ideas and methods, and its teaching will have a big effect in many directions. It is no new system; like almost all wisdom, it comes from the East. The old Hindu and Persian philosophers knew and practised it thousands of years ago, and got results which we have not approached as yet. The foundation is concentrated attention: we talk about giving attention to a thing, but we most of us do it in a very perfunctory, superficial way. Kipling's *Kim*, which is full of old, out-of-the-way, unsuspected philosophy, gives us a wonderful picture of concentrated attention in the contest of observation and memory between Kim and the native boy in Lurgan Sahib's shop. We talk admiringly of a man with "singleness of purpose," but what is that but concentrated attention? We admire him because he achieves his object, he gets there, while the ordinary man with half-a-dozen half-concentrated purposes wanders down all sorts of side ways and seldom arrives anywhere. In this world of keen competition and of little leisure, waste of brain-power leads inevitably towards disease and failure, while economy leads to health and attainment.

Beyond the question of the nature of our mental work lies the thing worked for—the object: the importance of this from the health and disease standpoint is very great. If the object is a worthy one, something of use, something elevating, not only for ourselves but for others, attainment will bring happiness and peace, and so tend towards a healthy condition of brain. If, on the other hand, we strive for an ignoble, selfish object, one that is almost certain to harm others, attainment will be like Dead Sea fruit in our mouths, dry and unsatisfying; there will be in us no sense of joy, and the injury we have done to others will recoil on ourselves, bringing disappointment and sorrow. These are they who walk in slippery places, while the unthinking world, seeing not their end, regards them with envy.

By learning to use our minds wisely, intelligently and economically, we shall accomplish far more and with far less expenditure of force; we shall grow in the true philosophical love of a high and simple life. The false aims, the conflicts, the disharmonies of this world will pass us by and leave us at peace.

Arterio-sclerosis, then, except in the sad cases of heredity, is very much of our own making; it almost comes under the heading of “an occupation disease”; it is a natural result of non-natural conditions of life. In the words of Isaiah: “Wherefore do ye spend money for that which is not bread, and your labour for that which satisfieth not?”

Writers have for descriptive purposes, and in most cases rightly, divided arterio-sclerosis into the two main types that I have described on p. 50—the hyperpiesis type and the atheromatous; but in general everyday practice we shall often come across cases that fall under neither head. Raised blood pressure, even to a considerable degree, 170 mm. to 180 mm., is frequently a temporary affair only, brought about by temporary causes, such as grief, anger, excitement or injudicious feeding. An altered mental outlook or a dose of calomel may banish it quickly; these cases are functional only. The wiser ones get to recognize their symptoms and the causes: these symptoms are chiefly a worried state of mind, irritability about little things, a sense of brain hurry, without the power to accomplish, and there may be passing

vertigo and tinnitus. Such is the day of small things which should never be ignored, for the day of small errors unheeded may pass on into the long night of disease.

CHAPTER III

THE WAY OF SAFETY

The pre-sclerotic state of raised pressure may last a considerable time—two or three years—but this can never be relied upon; every attack means extra work for the heart, and uses up for a time the reserve power of that long-suffering organ. Some hearts will respond well to the call for many years, but others give out and dilate. Continued raised pressure means one of two things—increased resistance and partial obstruction in the small arteries and capillaries, or increased ventricular action and force. An example of the first, which is far the more common, is seen in sclerosis; of the second, in aortic regurgitation. In the one case the heart is overcoming resistance—in the other, compensating for a leak. In the first, the error or departure from the normal is in the pipes—in the second, in the pump; but as both are vital organs and not mere bits of mechanism, they in time react on each other and both become altered. This change is, if vitality be strong, in the direction of conservatism, of accommodation to the new order of things. It is compromise, which may be effectual for a long time, but which, like most compromises, comes to failure in the end. Occasionally we find the sclerosis attacking the heart muscle itself or its arteries; such cases are obviously more serious. This condition generally comes on more or less at the end of the disease, but in a few cases quite early. Probably many of the cases of true angina pectoris occurring in early middle life are of this nature, the coronaries being the chief point of attack.

Besides excess of work and unwise work there is another cause which seems to raise pressure, and that is absorption of ptomaines or of the products of imperfect digestion; the overstrain cause and the absorption cause very often act concurrently, for imperfect assimilation is often the result of the unwise life. We see this more often, perhaps, where the unwise

life is accompanied by overeating and injudicious alcoholic stimulation, even if this is not very excessive: such people rarely give themselves time to digest. The supply of nerve energy that is necessary for perfect digestion is turned off at the main long before its proper time. We doctors have a first-hand personal knowledge of this cause, for owing partly to the intensive nature of our work and partly to the exigent importunity of our patients, we more often practise the vices we deplore than the virtues we praise; this is one of the reasons that our profession is the shortest-lived of all.

It is not quite clear how this absorption causes hypertension, but it is probably through an effect on endocrine glands, chiefly perhaps on the suprarenal and secondly on the thyroid. McGarrison has shown how often Graves' disease is caused by faulty intestinal digestion, and by the colitis that ensues; this he thinks stimulates the thyroid to overactivity or to faulty secretion. The suprarenals are probably affected still more than the thyroid, and the hyperthyroidism may be due to the upset of the balance of power; for these two glands through the sympathetic act and react on one another. It is fairly established that in some cases of Graves' disease we get hypertension and in some hypotension, though both conditions vary a good deal. It may be that the hypotension cases are those that are accompanied by suprarenal deficiency and that benefit by suprarenal treatment, and I believe it to be so. This would explain the conflicting opinions as to the value of this treatment. It has certainly succeeded admirably in some cases.

The co-ordination of all endocrine actions is so intricate that we can only puzzle out the way by careful clinical observation and experiment. In sclerosis, as a rule, we get evidence of hyperadrenalism and more certainly of hypothyroidism. Victor Horsley considered premature senility to be a form of myxœdema, which of course means pronounced hypothyroidism.

Biedl says: "A special pathogenetic significance is ascribed to thyroid insufficiency in the changes which occur in later life, and which are included in the term 'cachexia of old age,' or senile degeneration. The foundation for the theory that old age results from the changes in the thyroid gland lies in the fact that in old age the thyroid becomes atrophied, its follicles shrink,

and retrogressive changes take place in the epithelial cells. This is reinforced by the fact that there is a profound analogy between the signs of advanced old age and myxœdema. The falling of the hair and the dropping out of the teeth, the dry and wrinkled skin, the lowered temperature, the diminished perspiration, the indolent digestion and consequent emaciation, the decrease of mental power and the diminution of the activity of the entire nervous system—these are all symptoms that characterize chronic myxœdema.”

From the old partially true adage, that a man is as old as his arteries, we must go a step further back and say that a man is as old as his thyroid. This statement is probably not a final one, but it gives us a very valuable pointer in the way of treatment. Arterio-sclerosis and arterial degeneration are the almost universal accompaniments of thyroid deficiency, and are in most cases the result of that condition.

There is no main or royal road to the successful management of sclerosis; every little cause must be thought out and righted if possible, but as a foundation nothing is so important, I think, as a clean, healthy intestine, which hitherto has been much neglected. A microscopical examination of the fæces will often show what is wrong, and also the direction and results of treatment.

If the condition is thoroughly bad I begin with this rather drastic regime—twenty-four hours’ absolute starvation in bed, a tenth of a grain of calomel every hour for twelve hours and as much hot water as the patient can comfortably drink. This has a wonderful effect, and one’s subjective feelings after it tell us that evil has gone out of us and that good has taken its place.

The treatment by bowel antiseptics has been generally very unsatisfactory. Beta naphthol helps, perhaps, but the most good comes from duodenal extract in some form. Two grains of the dried powder about two hours after each meal in pill or tablet act well. Carnrick’s Secretogen tablets and the much-advertized Jubol are good, and both depend on some extract from the

duodenal mucosa. The unpleasant flatulence and the fermentation in the bowel soon disappear, and with them the mucus that is often shed from the coats of the colon. Better assimilation of food goes hand-in-hand with improved health and vigour. Intestinal dyspepsia is more far-reaching in its effects than gastric dyspepsia, but is often ignored.

The questions of food and cooking need most careful consideration. In the affairs of the soul Christ taught “that which cometh out of the man, that defileth the man”; in the affairs of the body, that which goeth into a man, that defileth the man. Both teachings are true and both are purposeful exaggerations and in a sense untrue. Out of the same mouth proceed blessings and cursings and into the same mouth go good food and poisons. The sin lies sometimes in the nature, but more often perhaps in the excess. The word “temperance” has been made to apply far too much to the use of alcohol. As a fact, I believe, far more nowadays are intemperate in eating than in drinking. As the tale of our years increases and as our physical life gets less strenuous and active, we require less food, or at any rate a different kind of food; the strong nitrogenous foods, as meat, become less called for, and if taken beyond our needs become poisons. The imperfect products of assimilation have to be dealt with by our excretory organs, which become overworked and unable to meet the demands. The scavengers cannot clear the dust-heaps. Hence arise kidney disease, faulty liver action, and raised blood pressure. And yet, how many of us seek to restore failing strength by stronger food and more stimulation? In this futility, kind but ignorant friends aid and abet. Think on what a little food a baby lives and grows, and, at the other end of life, on what a simple frugal diet really old people thrive and work.

It is all very well to lay down rules for food, calculated in calories for strong working men and athletes, but in approaching old age every one must become more or less a lawgiver to himself; this especially, if there is any tendency to hyperpiesis, for any food in excess of the body’s needs becomes for the time a poison. What we can digest easily and what with difficulty must be largely a personal question, but the direction should

certainly be towards a non-meat regime. The difference of climate makes comparison between the vegetable feeders of the tropics and ourselves of uncertain value, but it is a lesson to see what the Scotch gillie can do on a diet that is mostly oatmeal, milk and cheese. The gluttonous man and the wine bibber seem to get a pleasure out of life that the sober men miss, but we doctors see the other side of the shield, the after-suffering and the failure: the broken law brings its own punishment and there is no escape.

The physiological process of digestion is rather elaborate, and, fortunately for us, somewhat elastic. It is not a question entirely of certain substances in a test-tube exposed to certain digestive agents. This, of course, is the foundation, the basic process, but there is much variation in the nature of the food, in the external conditions of season, of heat and cold, of youth and age, of greater or less physical or mental demands on the food digested, and lastly on the supplies of saliva, of pepsin, of hydrochloric acid and of duodenal and pancreatic secretions. These are all under the control of the autonomous nervous system, which in the stresses of life often breaks down and fails to function properly. In the full tide of youthful energy our digestions seem to respond effectually to many and unreasonable demands, but this is more apparent than real; the bills may be a long time before they are sent in, but they come. The recuperative power of youth is marvellous, but probably never quite complete. After forty years of age it fails more and more, and then bills come in more quickly, while in old age it becomes almost a question of cash on delivery. Realizing this we should, in our own interests, treat our digestive apparatus with deep respect, and not with the scant courtesy we generally show, for on its faithful working depend our life, our health, our happiness and our powers of work. The farmer in his feeding of cattle, the head of a racing stable in his feeding and training of horses is far more scientific and wise than we and most other men are in the management of ourselves. This is why we end our days so often prematurely, or in paralysis and pain.

To go into actual details of diet is a thankless task, and not very helpful to the mass of men; it is better, I think, to point out directions only. If there are

symptoms of hyperpiesis, the chief object is to arrange a diet that contains a minimum of protein (especially of animal protein) compatible with life and energy. Most of such cases are leading a life of no great physical strain and will do well on a diet calculated in calories, that contains:

Protein	90
Fat	75
Carbohydrate	330 = Calories 2,400.

If this is insufficient, add to the fat or carbohydrate rather than to the protein.

The following tables from Dr. Sprigg's excellent work classify the common foods according as their energy production is due to protein, carbohydrate or fat. It will be noticed that milk is the only food in the lists which contains a good proportion of all three. Cheese, Brazil nuts and bread contain a fair proportion of at least two of the requisite food-stuffs.

If we take, for instance, the figures of lean beef in the first and second column, viz.:

Lean beef	90 + 10 we get its value
Chicken	79 + 21 we get its value
Bread	13 + 6 + 81 we get its value
Cheese	2 + 7 + 2 we get its value

By roughly arranging foods according to such a table we can arrive approximately at a suitable balance of its various constituents.

PERCENTAGE OF TOTAL HEAT VALUE OF FOOD FURNISHED BY ITS

PROTEIN		FAT		CARBOHYDRATE	
	Per cent.		Per cent.		Per cent.
Lean beef (boiled)	90	Butter	99	Tapioca (cooked)	98
Chicken	79	Bacon	94	Prunes (dried)	97
Mackerel	50	Cream	87	Figs (dried)	95
Skim Milk	37	Brazil Nuts	86	Rice (boiled)	89
Eggs	32	Fat Ham	81	Oysters	89
Beef with fat	25	Fat Beef	75	Potatoes (boiled)	88
Cheese	25	Cheese	73	Bread	81
Fat Ham	19	Eggs	68	Peas	72
Milk	19	Boiled Mutton	65	Milk	29
Bread	13	Milk	52	Cream	8
Potatoes	11	Mackerel	50	Brazil Nuts	4
Boiled Rice	10	Chicken	21	Cheese	2
Brazil Nuts	10	Boiled lean Beef	10		
Bacon	6	Bread	6		
Cream	5	Bananas	5		
Bananas	5	Potatoes	1		
Butter	5				

But the chemical side of the problem is not the only one to study. Idiosyncrasy, appetite for certain foods, distaste for others, and personal experience must all receive due consideration. The appetite for certain foods varies much at different periods of life. Fat is abhorrent to many children, and their lives and digestions are much troubled by the parental, but unscientific order to "clear up their plates." In later years the same persons will welcome fat and need it. Sugar again is loved by children, scorned in middle age, when its place is often supplied by wine or beer, and is again welcomed and of great value in old age, for it helps much to

maintain the failing heat of the body, to keep the home fires burning. Good cane-sugar at all stages of life seems to act as a special food, or at any rate stimulant, to the heart muscle; given in large quantities it often gives a failing heart a new lease of life. The digestibility of food is not governed entirely by its chemical composition. Twice cooked meat must contain nearly as much potential nourishment as once cooked, but experience tells most of us that we digest it with difficulty; and if the digestion of a food is not perfect, there is more unwholesome waste product to be cleared away; there is impure blood circulating in all our organs, but we feel it perhaps more in our brains; the lethargy, the depression, the irritability that we all recognize as the accompaniments of indigestion, all stand as cause and effect. Again hot fat and cold fat must contain the same constituents, but many can do well with cold fat, while they get bilious with the other. The same thing applies to butter. All these things we must honestly observe, each man for himself, and we must not let our likes and our appetites, nor our prejudices, sway the conclusions of personal experience and scientific knowledge. Dr. Haig has shown us how many of our ailments are connected with excess of uric acid in the blood, and how much this excess is due to certain foods such as meat, which contains what Fischer named purine bodies. The following extract from Halliburton's *Handbook of Physiology* will partially explain this subject:

Origin of Uric Acid.—Uric acid is not made by the kidneys; when these organs are removed, uric acid continues to be formed, and accumulates in the organs, especially in the liver and spleen. After extirpation of the liver in birds (in which animals uric acid is such an important katabolite) the formation of uric acid practically ceases, and its place is taken by ammonia and lactic acid, and the conclusion is therefore drawn that in these animals ammonia and lactic acid are normally synthesized in the liver to form uric acid. But in mammals this is not the history of uric acid formation; in these animals, including man, uric acid is the end-product of the metabolism of nuclein, from the bases of which it arises by oxidization. Nuclein, the main constituent of the nuclei of cells, yields on decomposition certain products

called purine substances, and their close relationship to uric acid is shown by their formulæ:

PURINE BASES.

Purine	$C_5H_4N_4$
Hypoxanthine (monoxypurine)	$C_5H_4N_4O$
Xanthine (dioxypurine)	$C_5H_4N_4O_2$
Adenine (amino-purine)	$C_5H_3N_4NH_2$
Guanine (amino-oxypurine)	$C_5H_3N_4ONH_2$
Uric acid (trioxypurine)	$C_5H_4N_4O_3$

“Just as the ordinary protein metabolism is both exogenous and endogenous, so is it the case with nuclein metabolism. There are certain kinds of food (such as liver and sweetbread) which are rich in nuclei, and others, such as meat, which are rich in purine bases (especially hypoxanthine). The increase in uric acid excretion after partaking of such food is exogenous, and those liable to uric acid disorders should avoid such articles of diet. Other forms of diet lead to an increase of uric acid formation by increasing the number of leucocytes in the blood, and there is a consequent increase in the metabolism of their nuclei. Increase in leucocytes may, however, be present independently of diet, and in the disease known as leucocythæmia this occurs to a marked degree; in such cases uric acid formation increases. Although special attention has been directed to the nuclei of leucocytes, because these can be readily examined during life, it must be remembered that the nuclein metabolism of all cells may contribute to uric acid formation. Uric acid, which originates by metabolism, is spoken of as endogenous.”

These purine bodies taken in excess, it is almost certain, tend to raise blood-pressure and to favour the production of gout and arterio-sclerosis. Much can be done, when these threaten, by reducing the amount of these in our food. There is, except in very extreme cases, no need for an absolute purine-free diet, but the diet should be in that direction and so towards a sensible, liberal vegetarianism. The following table of ordinary foods, showing the

purine contents in grains per pound, pint or teacup, is from the *Lancet*, 1906, vol. ii, p. 933, Potts:

Sweetbread	70.43	Halibut	7.14
Liver	19.26	Plaice	5.56
Beef Steak	14.45	Cod	4.07
Salmon	9.13	Beans	4.16
Chicken	9.06	Lentils	4.16
Loin of Pork	8.48	Oatmeal	3.45
Veal	8.13	Coffee	1.70
Ham	8.08	Ceylon Tea	1.21
Mutton	6.75	China Tea	0.75

In sclerosis perhaps the chief thing is to avoid the flesh extracts, such as beef-tea, strong meat soups, and rich gravies; for this reason boiled or stewed meats are better than fried or roast. This applies to fish and chicken also; a plain grill is good, but the frying-pan is a danger. Vegetable soups made with bone stock may be taken. Cheese, eggs and milk should supply the greater part of the nitrogenous or protein food. The better sorts of vegetable foods, such as oatmeal, lentils, peas and nuts, will all help to take the place of albuminous animal foods. A certain amount of fat should be taken. A fish and chicken diet contains too little fat and should be perfected by bacon, hot or cold. All the farinaceous foods are good, but the more starchy ones, such as sago, rice and cornflour, are with some people more liable to cause fermentative dyspepsia.

As age advances and our teeth fail in power we are apt to slip into the way of eating chiefly soft, well-cooked foods, in which what are known as vitamins are almost destroyed, but they are as essential as ever, and some uncooked fruit, salads and unboiled milk should enter into our diet.

When one thinks over the luxurious habits of our predecessors, the enormous amount of food, in excess of body needs, taken at big dinners, one must realize how we and they sinned against the light. The great war has taught most of us, with the exception perhaps of the new rich, how little we can live on and work on with complete efficiency. Economy of food,

simplicity of life, mean economy of waste and of expenditure of vital force, and our bitter war experiences should prove to be the path to longer lives, and to lives with a higher standard and greater results.

The question of alcohol in sclerosis is difficult, but very important. In the first place it depends somewhat on what a man has been accustomed to take. If he has taken a moderate amount with his food during the greater part of his life it does not always do to stop it. Alcohol is not a pressure raiser, as many think, but rather the reverse, and, if stopped suddenly, the digestive powers may be lessened and the tone of the body depressed. Lauder Brunton says: "All the Alcohols tend to dilate vessels, to diminish blood-pressure, and ultimately to diminish activity of the nervous tissues, although at first sight they may seem to have a stimulant action."

Alcohol contains so little nutritious food (with the exception of the sugars and extractives of wine and beers) that in itself it may be said to give nothing to the body; but there is something a bit beyond this—the bitter of the hop and the flavour, the bouquet of the grape have a stimulating effect on the nerves of taste and smell, and so produce, with some people, a better flow of saliva and gastric juice. This probably is how wine and beer in moderate quantities help digestion and appetite. There is still further the effect on the general nervous system. Alcohol without doubt enables a man to draw on his reserve of force, though it probably does not increase those reserves. This is, of course, drawing on one's capital, but the ability to do this may be of great use in emergencies, for it can help a man to live over evil times. But it must not be forgotten that such calls, if too large or often repeated, must in the end empty the reservoirs and leave him powerless and defenceless.

This question of alcohol must be regarded by physicians with common sense backed by scientific knowledge. The world is never going to be cured of alcoholic intemperance by intemperance in language and inaccuracy in thought.

Before I go into the more detailed treatment of sclerosis, I must just touch on the question of tobacco. Nicotine, without doubt, is one of the most powerful raisers of arterial pressure known, but in ordinary smoking very little gets into the system. Tobacco chewing and snuff-taking, both nearly extinct, probably introduce more nicotine into the body. The combustion in cigar smoking is so nearly complete that very little poison remains except on the lips. In ordinary cigarette smoking the combustion is about as complete as in the case of cigars, but, if the smoke is habitually inhaled, the absorption through the bronchial mucous membrane is much greater and, I think, must have an appreciable effect on blood-pressure. As in alcohol, so in tobacco, we must not lose sight of the effect on the general nervous system: on many people tobacco, in moderation, has a distinctly soothing influence, and so leads to a quietude of mind which compensates for any other small evil effect. Used in excess, it certainly weakens both nerve and will power, and often heart power. The first effect in experimental nicotine poisoning is greatly raised blood pressure, but at the end dangerously low pressure, and so we find that the ultimate effect of excessive smoking is a feeble, low-tension pulse that is often irregular. Oliver found that, in ordinary people who were not excessive smokers, tobacco raised the systolic pressure 10 to 15 mm., but that the diastolic was unaffected; the variations between the two thus becoming abnormal. This effect generally subsides in a quarter of an hour after smoking is finished.

It follows, I think, that a man with sclerosis should think this out for himself, for he is not in a position to take any unnecessary chances. Of another of our daily habits, it is wise to remember that coffee raises pressure more than tea, and China tea less than Indian.

There is no doubt that an indoor, sedentary life tends to high pressure far more than an active, outdoor one. This is partly due to the mental work and concentration that goes with it and partly to oxygen deficiency. The inhalation of oxygen lessens the viscosity of the blood and consequently the work of the heart and of the arterial muscles, while an excess of carbon-dioxide in the air increases the viscosity and the necessary heart force that

must be expended. The amount of oxygen we inhale in the open is necessarily more than in confined buildings, and we all feel how difficult it is to drive our brains in a close, stuffy atmosphere. This points to the economic value of ventilation and space. Wise, moderate exercise also tends to lower pressure by promoting a general quickening of circulation throughout the whole body; the stagnant pools are flushed out and oxygenation becomes more perfect. Even severe exercise in health only raises pressure temporarily; the subsequent level in health is below the original. But where there is evidence of cardio and arterial disease, exercise must be taken wisely, moderately, and only slowly increased. This is done very effectually in some health resorts, where the patient is trained to mount gradually increasing slopes. The same effect is reached by graduated exercises. In all these methods pulse rate and arterial pressure are tested and serve as guides to further advance. Certain Bath treatments, such as Nauheim and its English adaptations, help materially to lower pressure and to hasten a sluggish blood flow. These act chiefly by dilating the blood-vessels at the surface of the whole body and thus relieving internal congestions and increasing oxygen exchange.

The nature of our games and exercise in sclerosis needs careful thought. Golf, for instance, is very good if taken quietly, but some courses are so hilly that there is a good deal of strain put on the heart. If one is compelled to play on such a course, one must take it slowly, and not allow the spirit of the game to run away with one. Quiet sculling is one of the best exercises, for it works all the muscles of the body slowly and symmetrically, and the strain on heart and lungs is entirely under one's own control. Swimming, again, is a most healthy exercise, if taken quietly and not in very cold water. I know of nothing that improves one's wind so quickly. But all these exercises must be taken under advice and never pushed to the point of real fatigue or exhaustion.

The need of rest is as great as the need of the right sort of exercise. Late hours should be absolutely avoided and little brain work should be done in the evening, so that the brain and all its important arteries can sink into

relaxation and sleep at once. The rest to the heart and circulation is of course never so prolonged, but nature has designed them to get their rest only in moments. The heart in an adult rests more than thirteen hours out of the twenty-four, under quiet normal conditions, the time of rest being the diastole, and the time of work being the systole or contraction. There are few healthy men who could not walk a thousand miles in six weeks, walking about eight hours a day and resting for the remainder of their time; but there are not many men who could walk a thousand miles in a thousand consecutive hours, as the late Captain Barclay did, because the frequent interruptions to their sleep would exhaust them completely. In the same way, if the heart is forced to beat more quickly, it becomes sooner exhausted, for the extra work is taken from the diastole, the sleeping or resting time of the heart.

The blood pressure is generally ten degrees lower in the morning after a good night's rest; and for this reason, in severe cases, I often advise people to take a whole day in bed once a week, so that this lower pressure may be prolonged for twenty-four hours.

The non-medicinal treatment of sclerosis, prophylactic and curative, consists then chiefly in a philosophical, calm mode of life, in a firm restraint of passions, of worry and excitements, in a wise and economic use of brain power, in great moderation in eating and drinking, in giving as much of our time as is possible to fresh air and moderate, unhurried exercise, and lastly to the avoidance of all severe strains on mind or body. Very few of us are able to accomplish all these positive and negative virtues, but as far as in us lies, they should be our rule of life.

CHAPTER IV

THE MEDICINAL TREATMENT OF SCLEROSIS

The question now arises, what can the Physician, by real therapeutics, do to help? Much, I feel sure, if he will carefully and courageously apply the knowledge that has come to us of the late years. Firstly, Lauder Brunton showed us how we can quickly reduce the blood pressure that is one of the chief symptoms in angina pectoris, the immediate effect of amyl nitrite in reducing arterial tension, and the slower but very valuable effects of nitroglycerin and the other nitrites. This has enabled us to meet many an emergency, but the effects of these are not sustained enough to become a steady daily treatment. The orthodox remedy has been for years Iodide of Potassium; this has been by the teachers of the past so driven into us that, in spite of failures that far outweigh the successes, it has become heterodox to doubt it. Yet one must confess that one has been for years prescribing it more from a sense of duty than from any confident hope of its results. In a few cases, where there is cardiac pain and symptoms of new or chronic aortitis, it certainly does good, but in the average case of hyperpiesis it is a very poor reed to trust.

How does Iodide act? Chiefly, in all probability, by stimulating the thyroid to pour more of its secretion into the blood stream. Thus is produced the absorption of gummata, of new lowly formed tissues and the thickenings of fibrositis and rheumatism. We see the same result more decidedly and more constantly in the treatment of myxœdema by thyroid feeding.

As I have shown before, senile degeneration, premature and normal, is largely due to general endocrine degeneration and deficiency, and it has many relations to myxœdema. Thyroid deficiency as a rule means raised blood pressure, and, as has been shown in young animals deprived of their

thyroids, a strong tendency to get atheromatous disease of the large arteries, even in youth.

The parallels are many, and so are the lessons they convey. If one had a perfectly functioning thyroid in arterio-sclerosis, one might with some confidence treat it by Iodides; but as we know that such is not the condition in the great majority of cases, we should expect to fail. One would have to give Iodide in enormous quantities with any hope to whip up a senile thyroid into effectual activity, and then in all probability fail. (This may be the explanation of the miracles worked by big doses of Iodide in syphilitic gummata in the brain, but in these cases there is rarely evidence of thyroid deficiency.)

Why then do we take the long and disagreeable route to our object by giving Iodide? Thyroid feeding acts here just as it does in myxœdema. We have to take into consideration the comparative failure of the other endocrine glands, but that only adds to the charm and attractiveness of the problem. The suprarenal and the pituitary are the most important, but one must not neglect the sexual glands. I feel strongly that thyroid treatment alone or in combination is the rational method to adopt in sclerosis: personal experience confirms this every day. We purposely side-track the worn-out thyroid of sclerosis and senility, and, by thyroid feeding, restore to a great extent the whole endocrine harmony, thus giving a new tone and stimulus to existence. We shall have to use more or less thyroid feeding for the remainder of life, but we must show our patients that it is not a medicine but a necessary food. As in myxœdema, so in sclerosis, thyroid feeding is the only means, as yet known, of producing absorption of those thickenings that characterize both diseases.

I have had lately the opportunity of watching and treating a thorough case of myxœdema in a lady about forty years of age, a very intelligent and educated woman. She weighs her symptoms and knows exactly when she needs more or less thyroid; she judges by her head feelings, vertigo, the sense of fatigue, etc., and by her hair falling out. But thyroid in sufficient doses produces a quick, irritable heart action; this she can quiet by taking

suprarenal extract. Under sufficient thyroid her blood pressure comes down to normal and the suprarenal does not raise it. With this balanced combination she can lead a useful and enjoyable life. The same thing happens more or less in later life, where there is high pressure and other signs of arterial thickening.

Suprarenal by the mouth seems to me to have the power of raising low pressures to the normal—a most valuable property in the asthenia of pneumonia and diphtheria—but there comes in some check action which prevents its causing hypertension.

Dr. Sajous, the well-known scientific physician of America, at the annual meeting of the American Therapeutic Society in May 1920, said: “With regard to the adrenals, in his opinion, clinical experience with the human subject did not bear out the statements of physiologists that therapeutic doses of adrenalin had any inhibitory effects on the functions of the gastrointestinal canal; in fact, the effect was quite the opposite. The supposed blood-pressure raising power of adrenalin also proved misleading; indeed, one of his workers found that, if there was one thing that adrenalin did not do, it was to raise blood pressure.”

In this direction we must realize the wonderful selective action of adrenalin, that of dilating the coronary arteries, contracting the arteries and arterioles of the skin and of the great splanchnic area and of dilating the deep arteries of the limbs. These selective actions explain the great relief it gives in asthma and acute respiratory distress, such as œdema of the lungs, where it bleeds a man into his own distal vessels.

The absence of raised pressure in the brachial and radial arteries may be thus explained also. Again we must not forget that there may be, and probably is, something in the extract of the whole gland besides adrenalin. Administered by stomach the extract has, I think, far more effect than pure adrenalin given in the same way, and more lasting. Given by rectum, adrenalin acts quickly and powerfully. In France thyroid treatment is

sometimes carried out by injections of an extract of the whole gland subcutaneously, but it is difficult to get an extract that is stable.

In persons of middle age it is far more common to get the unpleasant heart effects of thyroid than in really old people, but it is not by any means an absolute rule. I know of two women well over ninety who cannot on this account take thyroid by itself for long. After sixty, thyroid in combination with pitglandin (the dried extract of the pars-anterior of the hypophysis) becomes very useful. At this time of life the internal secretion of the sexual glands begins to fail, and pitglandin, I think, compensates for their deficiency. When one thinks of the extraordinary influence that the pars-anterior has in youth on growth, and non-growth, causing gigantism and infantilism, one cannot be surprised at its having some powerful influence at the other end of life. It can be given in large doses, 1 or 2 grains, and seems to have no poisonous effects. Unlike pituitrin (pars-posterior), it does not raise blood pressure, but, I believe, has the opposite effect. The secretions of testis and ovary are probably mild pressure raisers, and as they enter into the formation of nearly all the pluri-glandular preparations that are on the market, this should be recognized. What may be of great value in neurasthenia, where the pressure is generally sub-normal, may be harmful in sclerosis. Most of these preparations contain also whole pituitary, and this again is harmful in plus pressure. It always strikes me as unscientific to give whole pituitary in any case, for the two parts are largely antagonistic in their action. The extracts of the sexual glands have been given, with the object of restoring sexual power (except for its side effects a thing of doubtful value), but if they prove to be pressure raisers, sclerosis will do better without them. In all probability pitglandin has a stimulating effect on the gonads, as it has on sexual development in childhood, but it does not raise pressure.

We must regard it as a clinical fact that the thyroid reduces pressure. The original experiments of Oliver and Schäfer seem conclusive. Brunton said: "Thyroid gland, when taken by the mouth, dilates the peripheral vessels, making the skin warm and moist, and quickens the pulse. In this respect it

antagonizes the suprarenal secretion. Besides this effect on blood pressure, it has other effects on metabolism which are important." Biedl says: "If thyroid or iodothyrim be given continuously for two or three weeks, the amount of CO-excretion will be increased by 15 to 16 per cent. The nitrogenous interchanges are invariably disturbed by thyroid; the increased decomposition of albumen is expressed by an increased excretion of nitrogen. Thyroidism also brings about a considerable increase in the amount of calcium excreted in the fæces, the calcium carrying off with it a large proportion of phosphorus." This phosphorus loss may explain some of the feeling of weakness and exhaustion that thyroid feeding often causes; this a good phosphate food such as Bynogen will help to remove.

It is thus evident that we have in thyroid medication something much more than a mere tension depressor. Its other properties, influencing excretion, explain to some extent its sphere of usefulness and its drawbacks. The increased excretion of calcium may be very beneficial in atheroma, but is probably not an advantage in other ways. An observant man, who did not need it, was taking thyroid rather freely; he cut his finger and could not get it to heal. Thinking that this might be the result of calcium loss, he stopped the thyroid, and his wound healed, as he said, in a few hours. To the increased general metabolism that thyroid produces is due its striking effects in myxœdema: the new tissues, abnormal in quantity if not in essence, are absorbed, and the face and limbs resume their natural appearance. Senility and myxœdema are so closely related that one would naturally expect the same results to ensue. A fair steady trial will prove that such is the case, but all such clinical experiments need to be carried out in each individual case with thoroughness, faith and intelligent observation. Failure may at first be the verdict, but it is in most cases due to some defect in method. A thoughtful review of our physiological knowledge of endocrine actions will often show where lies the fault. A carping critic may say, What is your physiological knowledge of the action of these glands? And one must humbly answer, Not very much. But we are going to find our way through the twilight of our ignorance, partly by physiological

experiment, but chiefly by clinical results. We must also be encouraged by the thought that all these clinical trials are without risk. What matters it if we produce temporary symptoms of hyperthyroidism? We can always correct them or retrace our steps. Russell Lowell's proverb was, "A man who never makes a mistake never makes anything." In such therapeutics we shall discover the pathways of natural curative medicine. It is not the introduction into human bodies of vegetable and mineral remedies or poisons; it is simply the introduction of correlated gland extracts from other mammals. As far as we know, the glands of the herbivora answer as well as the glands of the omnivora, but we may have to correct this view.

In treating hyperpiesis, we shall come across some cases that seem to yield to thyroid treatment hardly at all, and here the trouble is often found to be chiefly, if not entirely, in intestinal absorption of deleterious products of faulty digestion; there may or may not be signs of colitis; but, as I have said before, duodenal extract freely given alters the condition rapidly in most cases. In these cases probably the absorption of these residues of defective digestion produces hyperadrenia, and the adrenal-thyroid balance is disturbed. Dr. Sajous (in his paper previously mentioned) claims that calomel in very small doses, not exceeding gr. $\frac{1}{12}$, is a powerful activator of the thyroid secretion; he claims that so given it has been effective in checking epidemics of diphtheria. It probably acts partly as an intestinal bactericide and partly by its action on the thyroid, increasing the defensive anti-bodies in the blood.

There is one other medicine that has a very decided action on some forms of raised pressure, and that is hippuric acid. This again is a natural or endogenous cure. It was introduced by the late Dr. George Oliver of Harrogate, and is chiefly used as sodium or lithium hippurate. For some time before his death Dr. Oliver was seeking for a reasonable explanation of its action, but his death unfortunately came before he had carried this through. I can give no theory, but of its effects in many cases I feel convinced, though I cannot say beforehand in which class of case it will

answer. In the pre-sclerotic cases of middle life it often acts quickly and well, even better than thyroid. Again, in old age, in cases where one would expect thyroid to give the better results, it answers equally well.

Two of my patients—old ladies well over seventy—have had hyperpiesis for twelve years or more; they were condemned by good men many years ago. They have lived on hippurate almost without breaks all these years, and it has kept their pressure down to a reasonable working-point. With both I have tried thyroid treatment with very slight success; they always come back to their hippurate as the well-tried friend. I have another patient aged ninety-four, who has had periods of high pressure for twenty years, and if she misses her daily dose the pressure goes up and her head gets giddy and confused. It is a very wholesome medicine, causes no gastric disturbance, and is easily soluble; from 5 to 15 grains a day are generally needed. If there are gouty symptoms I give the lithia salt; if not, the sodium. The benzoates have, I believe, much the same action, but they are not easily digested and large doses are required. In bad and obstinate cases the hippurate and thyroid treatment sometimes act well together.

I have written now all that I know from personal experience about the treatment of this complicated condition we call arterio-sclerosis; and many of the symptoms and much of the treatment I have been able to watch and verify for myself by almost hourly contact with it. Nevertheless, this treatment will have to stand the racket and assaults of both friendly and hostile critics, and so it should be. In the practice of medicine we must be prepared to maintain our position and to give a reason for the faith that is in us, for the issues are so great and affect others rather than ourselves.

I have said that this is somewhat of an occupation disease: it is a vicious circle, not originating necessarily in any vice, but in the pressure of life and the stress of circumstances. The causes often we cannot remove entirely, but only modify, and so we are compelled to treat symptoms only; our ambition and longing will be to do much more than this—to bring about a perfect reform; but in the management of this, as in most morbid conditions, compromise, with as much true wisdom as we can squeeze into it, is the

only practical policy. Our endeavour must be to break the vicious circle whenever and wherever we can. The lowering of pressure does not sound a great thing in itself, but it immediately lessens the work of the heart, sets more free the circulation of blood in the brain, often restores the power of sleep and so ensures comparative rest all round. It thus gives us a breathing space in which we can consider and attack the origins of evil.

In conclusion I ask my readers to consider well Sir Clifford Allbutt's most wise words on this question: "In entering upon the discussion of vasodilatation, as contrasted with agents which bring about this change indirectly by modifying the causes of morbid constriction, we have to consider how far mere dilatation—brought about, that is to say, immediately and singly—serves any good purpose.

"We are told that to act thus directly upon the vessels is but to treat a symptom, and is therefore absurd. But whatsoever be our judgment on this or any such particular effect, the common denunciation of treating symptoms, which sounds very philosophical, is surely but a parrot phrase. Why should we not treat a symptom? If in granular kidney by mere pressure reduction the grievous headache be abated, or in angina pectoris the pain be thus charmed away, we have so far at any rate a substantial gain.

"In renal diseases it is generally agreed that, on the whole, with due caution, to lower pressure is helpful. Moreover, if by mitigation of his suffering the patient gets a chance of picking up in many other ways, are we not more than justified in our interference, narrow as it may seem? We never know what interference may cut a link in a vicious circle. If we cannot stop the crack in the water-pipe, we need not throw away the mop.

"The warning should run not against the treatment of a 'mere symptom,' but lest, while giving our attention to the symptom and snatching at an immediate advantage, we lose our grip of the case as a whole."

In whatever treatment we adopt for sclerosis, we must bear in mind that, under the raised pressure, the heart is very apt to dilate. To lower pressure alone in this condition often ends in failure. We must brace up the heart at

the same time as we lower resistance. For the purpose strophanthus in large doses and strychnia are the best drugs. Digitalis is occasionally necessary, but more often does harm than good. Given, however, in this well-tried old form it contracts the ventricle without raising pressure:

Rx	Pulv. Digitalis	gr. 1–2
	Pil. Hydrarg.	gr. 1
	Pil. Scillæ Co.	gr. $1\frac{1}{2}$
	Ft. pil. to be taken twice a day.	

The compound gland tablet that I chiefly use has this formula:

Rx	Thyroid Extract dried	gr. $\frac{1}{2}$
	Orchitic Extract dried	gr. $\frac{1}{2}$
	Ovarian Extract dried	gr. $\frac{1}{2}$
	Anterior Pituitary dried	gr. $\frac{1}{2}$
	Ft. tab. 1.	

This can be taken three times daily and acts best, I think, if taken before food. The amount of thyroid can be doubled if necessary. In cases where there is much prostration the anterior pituitary can be increased to 2 grains. The orchitic and the ovarian can be omitted if thought advisable.

Hitherto many of us have looked on this condition, arterio-sclerosis, as a tenant's fixture, to be got rid of only when he moves into his next habitation. This is neither fair to ourselves nor to our patients. It is a hopeless stultification of our more recent physiological knowledge and discoveries. It is a pessimistic faith that deserves failure and gets it. It is a silent denial of those healing powers which Christ so lovingly displayed when on earth, and which are still the heritage of those who can see and believe.

Printed in Great Britain by
UNWIN BROTHERS, LIMITED
WOKING AND LONDON

THE ROAD TO A HEALTHY OLD AGE

By T. BODLEY SCOTT, M.R.C.S. (Eng.), Author of "Why Do We Die?"
etc. Second Edition. Cloth, 5s. net.

"Written in a popular style, the book has been prepared alike for doctors and laymen, and deals in a common-sense way with the subjects of prolongation of life; conservation of health, and the habits necessary for its promotion; food and food values; nature and treatment of premature senility, especially arterio-sclerosis or thickening of the arteries; thyroid treatment; and chronic bronchitis and asthma."—*Chambers's Journal*.

"We have perused this book with pleasure, and cordially recommend it to our readers."—*Medical Times*.

"We can cordially commend this work to readers of advancing years, whether medical or lay. To the former it presents an old subject in a new way, and to the latter will convey some very wholesome material for thought."—*British Medical Journal*.

T. FISHER UNWIN LTD, 1 ADELPHI TERRACE, LONDON, W.C. 2

INSTINCTS OF THE HERD IN PEACE AND WAR

By W. TROTTER. Revised and Enlarged. Fifth Impression. Large Crown
8vo, cloth. 8s. 6d. net.

"An exceedingly original essay on individual and social psychology."—*The New Statesman*.

"It is a balanced and inspiring study of one of the prime factors of human advance."—*The Times*.

"The main purpose of Mr. Trotter's book, which may be recommended both for its logic and its circumspection, is to suggest that the science of psychology is not a mass of dreary and indefinite generalities, but if studied in relation to other branches of biology, a guide in the actual affairs of life, enabling the human mind to foretell the course of human action."—*Daily Telegraph*.

T. FISHER UNWIN LTD, 1 ADELPHI TERRACE, LONDON, W.C. 2

A HANDBOOK FOR WOMEN **WOMAN AND MARRIAGE**

By MARGARET STEPHENS With a Preface by Dr. MARY SCHARLIEB
and an Introduction by Mrs. S. A. BARNETT (6th Impression.) Cloth, 6/-
net (post free 6/6).

“‘Woman and Marriage’ is an outspoken book which should be carefully read by those for whom it was written. It is not a book for boys and girls; it is a physiological handbook, thoroughly well written, orderly, wholesome, and practical. We recommend this work to all who want a full account in simple words of the physical facts of married life. All the difficulties of the subject are handled fearlessly, gravely, and reverently in this book, and as it must be kept out of the reach of mere curiosity, so it deserves thoughtful study by those of us whose lives it touches.”—*The Spectator*.

“If more such books were written, and more such knowledge disseminated, it would be a good thing for the wives and mothers of the present day.”—*The Times*.

T. FISHER UNWIN LTD, 1 ADELPHI TERRACE, LONDON, W.C. 2 *and of all Booksellers.*

PSYCHOPATHOLOGY OF EVERYDAY LIFE

By PROFESSOR SIGMUND FREUD, LL.D. Authorised English Edition with
an Introduction by A. A. BRILL, PH.B., M.D. Sixth Impression. Demy 8vo,
cloth. 12s. 6d. net.

“This book is a symptom of a welcome tendency in the study of psychopathology, and provides, from one point of view, a very desirable reaction from the methods of Lombroso, Kraft-Ebing, and similar writers. . . . Professor Freud deserves the high praise and true gratitude of genuine scientists.”—*The Observer*.

“The book is extraordinarily interesting and full of hints for the student of human nature.”—*Nature*.

T. FISHER UNWIN LTD, 1 ADELPHI TERRACE, LONDON, W.C. 2

UNWIN'S "CHATS" SERIES

PRACTICAL HANDBOOKS FOR COLLECTORS

Most people nowadays are collectors in a small way of Autographs, China, Furniture, Prints, Miniatures, or Silver, and would take up these fascinating hobbies more extensively, and collect with profit, if they had a knowledge of the subject.

It is to the beginner and would-be collector that Unwin's "Chats" Series of practical handbooks especially appeal. They are the recognized standard guides to collecting, each volume being the work of an expert on the subject dealt with.

Each volume is profusely illustrated with carefully-chosen specimens of the various styles and periods.

Full Indices, Bibliographies, and Lists of Sale Prices at Public Auctions are included in the volumes.

"As this is the age of collectors, so it is the age of books for their guidance. Mr. Unwin's series of books for collectors now includes twenty-one volumes, and if bargains are missed it is certainly not the fault of the various writers." *The Nation*.

HOW TO COLLECT WITH PROFIT

is the keynote of the series. The phenomenal prices realized at auction sales are obtained by those who have collected wisely. Prices are still rising, and those who have the knowledge are buying for future rises. Ask always for and see that you get UNWIN'S "Chats" Series—the standard popular handbooks on collecting.

THE "CHATS" SERIES IS ON SALE AT ALL BOOKSELLERS THROUGHOUT THE WORLD AND IS PUBLISHED BY T. FISHER UNWIN LTD., 1 ADELPHI TERRACE, LONDON, W.C. 2

LIST OF VOLUMES

Chats on English China. By ARTHUR HAYDEN. Illustrated with reproductions of 156 marks and 89 specimens of china. Cloth, 12s. 6d. net.

Sixth Impression.

This is the standard work on the subject. The volume will enable the possessors of old china to determine the factories at which their ware was produced.

“It gives in a few chapters just what the beginner wants to know about the principal varieties of English ware. We can warmly commend the book to the china collector.” *Pall Mall Gazette*.

“So simply yet so thoroughly written, that it is a sage guide to the veriest tyro in china collecting.” *Bookman*.

Chats on Old Furniture. By ARTHUR HAYDEN. With a coloured frontispiece and 104 other Illustrations. Cloth, 12s. 6d. net. Fifth Impression.

“The hints to collectors are the best and clearest we have seen; so that altogether this is a model book of its kind.” *Athenæum*.

“A fully illustrated practical guide for collectors.” *The Times*.

“Mr. Hayden has worked at his subject on systematic lines, and has made his book what it purports to be—a practical guide for the collector.” *The Saturday Review*.

Chats on Old Prints. How to Collect and Identify. By ARTHUR HAYDEN. With a coloured frontispiece and 72 full-page plates. Cloth, 10s. 6d. net. Fifth Impression.

Every branch of the subject is carefully and explicitly handled in this book, and valuable information as to technical processes and identification of prints is given.

“If there is a better book of its kind on print collecting we have not yet come across it.” *Daily Graphic*.

“A very useful handbook for beginners, intended to help any reader of artistic tastes, but very moderate means, to collect to good purpose.” *The Times*.

Chats on Costume. By G. WOOLLISCROFT RHEAD, R.E. With a coloured frontispiece and 117 other Illustrations. Cloth, 10s. 6d. net. Second Impression.

A practical guide to historic dress. “Clothes” is a subject that has been neglected by collectors, and this book will be a useful guide to those who desire to repair that neglect by forming a collection.

“A book that is at once the work of an authority on the subject of costumes, and one that helps to enlarge our range of selection.” *Pall Mall Gazette*.

Chats on Old Miniatures. By J. J. FOSTER, F.S.A. With a coloured frontispiece and 116 other Illustrations. Cloth, 6s. net.

This book presents in a concise and popular form a variety of valuable information on the collection and preservation of miniatures, on the leading English and French artists, and on the specimens exhibited in public galleries.

“Mr. Foster is truly a guide, philosopher and friend. He tells us not only how to judge and how to buy miniatures, but how to take proper care of them. . . . The splendid photographs by which the book is enriched adds in a great measure to its attractiveness and utility.”
Aberdeen Free Press.

Chats on Old Lace and Needlework. By MRS. LOWES. With a frontispiece and 74 other Illustrations. Cloth, 10s. 6d. net. Third Impression.

Written by an expert and enthusiast in these most interesting branches of art. The low price at which the work is issued is exceptional in dealing with these subjects, and it is remarkable in view of the technical knowledge displayed and the many photographic illustrations which practically interleave the book.

“In commendable, clear and concise style Mrs. Lowes explains the technical features distinguishing each example, making the book the utmost value in identifying samples of old lace.”—*Weldon's Ladies' Jour*.

Chats on Oriental China. By J. F. BLACKER. With a coloured frontispiece and 70 other Illustrations. Cloth, 10s. 6d. net. Third Impression.

Will be of the utmost service to collectors and to all who may have old Chinese and Japanese porcelain in their possession. It deals with oriental china from the various standpoints of history, technique, age, marks and values, and is richly illustrated with admirable reproductions.

“A treatise that is so informing and comprehensive that it commands the prompt recognition of all who value the choice productions of the oriental artists. . . . The illustrations are numerous and invaluable to the attainment of expert knowledge, and the result is a handbook that is as indispensable as it is unique.” *Pall Mall Gazette*.

Chats on English Earthenware. A companion volume to “Chats on English China.” By ARTHUR HAYDEN. With a coloured frontispiece, 150 Illustrations and tables of over 200 illustrated marks. Cloth, 10s. 6d. net. Third Impression.

“To the ever-increasing number of collectors who are taking an interest in old English pottery . . . will be found one of the most delightful, as it is a practical work on a fascinating subject.” *Hearth and Home*.

“Here we have a handbook, written by a well-known authority, which gives in the concisest possible form all the information that the beginner in earthenware collecting is likely to need. Moreover, it contains one or two features that are not usually found in the multifarious “guides” that are produced to-day.” *Nation*.

Chats on Autographs. By A. M. BROADLEY. With 130 Illustrations. Cloth, 6s. net.

“Being an expert collector, Mr. Broadley not only discourses on the kinds of autograph he owns, but gives some excellent cautionary advice and a valuable ‘caveat emptor’ chapter for the benefit of other collectors.” *Westminster Gazette*.

“It is assuredly the best work of the kind yet given to the public; and supplies the intending collector with the various sources of information necessary to his equipment.” *Manchester Guardian*.

Chats on Pewter. By H. J. L. J. MASSÉ, M.A. With 52 half-tone and numerous other Illustrations. Cloth, 10s. 6d. net. Second Impression.

“It is a remarkably thorough and well-arranged guide to the subject, supplied with useful illustrations and with lists of pewterers and of their marks so complete as to make it a very complete and satisfactory book of reference.” *Manchester Guardian*.

“Before setting out to collect old pewter it would be as well to read Mr. Massé’s book, which is exhaustive in its information and its lists of pewterers, analytical index, and historical and technical chapters.” *Spectator*.

Chats on Postage Stamps. By FRED J. MELVILLE. With 57 half-tone and 17 line Illustrations. Cloth, 10s. 6d. net. Second Impression.

“The whole book, with its numerous illustrations of excellent quality, is a *vade mecum* for stamp collectors, even though their efforts may be but modest; we congratulate Mr. Melville on a remarkably good guide, which makes fascinating reading.” *Academy*.

“There is no doubt that Mr. Melville’s book fills a void. There is nothing exactly like it. Agreeably written in a popular style and adequately illustrated, it is certainly one of the best guides to philatelic knowledge that have yet been published.” *World*.

Chats on Old Jewellery and Trinkets. By MACIVER PERCIVAL. With nearly 800 Illustrations. Cloth, 6s. net.

“The book is very thorough, dealing as it does with classic, antique and modern ornaments; with gold, silver, steel and pinchbeck; with the precious stones, the commoner stones and imitation.”—*Outlook*.

“‘Chats on Old Jewellery and Trinkets’ is a book which will enable every woman to turn over her jewel-case with a fresh interest and a new intelligence; a practical guide for the humble but anxious collector. . . . A good glossary of technicalities and many excellent illustrations complete a valuable contribution to collector’s lore.” *Illustrated London News*.

Chats on Cottage and Farmhouse Furniture. A companion volume to “Chats on Old Furniture.” By ARTHUR HAYDEN. With a coloured frontispiece and 75 other Illustrations. Cloth, 10s. 6d. net. Second Impression.

“One gets very much for one’s money in this book. Seventy-three full-page illustrations in half-tone embellish a letterpress which is replete with wise description and valuable hints.” *Vanity Fair*.

“Mr. Hayden’s book is a guide to all sorts of desirable and simple furniture, from Stuart to Georgian, and it is a delight to read as well as a sure help to selection.” *Pall Mall Gazette*.

“Mr. Hayden writes lucidly and is careful and accurate in his statements; while the advice he gives to collectors is both sound and reasonable.” *Westminster Gazette*.

Chats on Old Coins. By FRED W. BURGESS. With a coloured frontispiece and 258 other Illustrations. Cloth, 10s. 6d. net. Second Impression.

“A most useful and instructive book . . . will prove a boon to the intending collector of old coins and tokens, and full of interest to every collector. As was to be expected of any volume of this series, the illustrations are numerous and good, and greatly assist the reader to grasp the essentials of the author’s descriptions.” *Outlook*.

“The author has not only produced ‘a practical guide for the collector’ but a handy book of reference for all. The volume is wonderfully cheap.” *Notes and Queries*.

Chats on Old Copper and Brass. By FRED W. BURGESS. With a coloured frontispiece and 86 other Illustrations. Cloth, 6s. net.

“Mr. F. W. Burgess is an expert on old copper and bronze, and in his book there is little information lacking which the most ardent collector might want.” *The Observer*.

“Italian bronzes, African charms, Chinese and Japanese enamels, bells, mortars, Indian idols, dials, candlesticks, and snuff boxes, all come in for their share of attention, and the

reader who has mastered Mr. Burgess's pages can face his rival in the auction-room or the dealer in his shop with little fear of suffering by the transaction." *The Nation*.

Chats on Household Curios. By FRED W. BURGESS. With 94 Illustrations. Cloth, 6s. net.

"Mr. Burgess gives much information about such attractive antiques as old glass and enamels, old leather work, old clocks and watches, old pipes, old seals, musical instruments, and even old samplers and children's toys. The book is, in short, an excellent and comprehensive guide for what one may call the general collector, that is, the collector who does not confine himself to one class of antique, but buys whatever he comes across in the curio line, provided that it is interesting and at moderate price." *Aberdeen Free Press*.

Chats on Japanese Prints. By ARTHUR DAVISON FICKE. With a coloured frontispiece and 56 Illustrations. Cloth, 6s. net. Third Impression.

"Mr. Ficke writes with the knowledge of the expert, and his history of Japanese printing from very early times and his criticism of the artists' work are wonderfully interesting." *Tatler*.

"This is one of the most delightful and notable members of an attractive series. . . . A beginner who shall have mastered and made thoroughly his own the beauty of line and the various subtlety and boldness of linear composition displayed in these sixty and odd photographs will have no mean foundation for further study."—*Notes and Queries*.

Chats on Old Clocks. By ARTHUR HAYDEN. With a frontispiece and 80 Illustrations. 2nd Ed. Cloth, 10s. 6d. net.

"A practical handbook dealing with the examples of old clocks likely to come under the observation of the collector. Charmingly written and illustrated." *Outlook*.

"One specially useful feature of the work is the prominence Mr. Hayden has given to the makers of clocks, dealing not only with those of London, but also those of the leading provincial towns. The lists he gives of the latter are highly valuable, as they are not to be found in any similar book. The volume is, as usual with this series, profusely illustrated, and may be recommended as a highly interesting and useful general guide to collectors of clocks." *The Connoisseur*.

Chats on Old Silver. By ARTHUR HAYDEN. With a frontispiece, 99 full-page Illustrations, and illustrated table of marks. Cloth, 10s. 6d. net. Third Impression.

"Mr. Hayden's 'Chats on Old Silver' deals very thoroughly with a popular branch of collecting. There are a hundred full-page illustrations together with illustrated tables and charts, and the student of this book can wander round the old curiosity shops of these islands with a valuable equipment of knowledge. . . . Altogether we have here a well-

written summary of everything that one could wish to know about this branch of collecting.” *The Sphere*.

“The information it gives will be of exceptional value at this time, when so many families will be forced to part with their treasures—and old silver is among the most precious possessions of the present day.” *Morning Post*.

Chats on Military Curios. By STANLEY C. JOHNSON, M.A., D.Sc. With a coloured frontispiece and 79 other Illustrations. Cloth, 6s. net.

“Mr. Johnson in this book describes many of the articles a collector should be on the look out for, giving short but informative notes on medals, helmet and cap badges, tunic buttons, armour, weapons of all kinds, medallions, autographs, original documents relating to Army work, military pictures and prints, newspaper cuttings, obsolete uniforms, crests, stamps, postmarks, memorial brasses, money and curios made by prisoners of war, while there is also an excellent biography on the subject. The author has, indeed, presented the reader with a capital working handbook, which should prove a friendly and reliable guide when he goes collecting.” *Field*.

Chats on Royal Copenhagen Porcelain. By ARTHUR HAYDEN. With a frontispiece, 56 full-page Illustrations and illustrated tables of marks. Cloth, 10s. 6d. net.

“This very beautiful and very valuable book will be eagerly welcomed by lovers of porcelain. . . . Mr. Hayden describes with great skill and preciseness all the quality and beauty of technique in which this porcelain excels; he loves it and understands it, and the examples he has chosen as illustrations are a valuable supplement to his descriptions.” *Bookman*.

A NEW “CHATS” VOLUME TO BE ISSUED IN THE AUTUMN
OF 1920. ORDER EARLY

Chats on Old Sheffield Plate. By ARTHUR HAYDEN. With frontispiece and 53 full-page Illustrations, together with makers’ marks. Cloth, about 10s. 6d. net.

Old plated ware has, by reason of its artistic excellence and its technique, deservedly won favour with collectors. The art of making plated ware, which originated at Sheffield (hence the name “Sheffield plate”), was continued at Birmingham and London, where a considerable amount of “old Sheffield plate” was made, in the manner of its first inventors, by welding

sheets of silver upon copper. The manufacture lasted roughly a hundred years. Its best period was from 1776 (American Declaration of Independence) to 1830 (Accession of William IV). The author shows reasons why this old Sheffield plate should be collected, and the volume is illustrated with many examples giving various styles and the development of the art, together with makers' marks. Candlesticks and candelabra, tea-caddies, sugar-baskets, salt-cellars, teapots, coffee-pots, salvers, spoons, and many other articles shown and described in the volume indicate the exquisite craftsmanship of the best period. The work stands as a companion volume to the author's "Chats on Old Silver," the standard practical guide to old English silver collecting.

Bye Paths in Curio Collecting. By ARTHUR HAYDEN, Author of "Chats on Old Silver," etc. With a frontispiece and 72 full-page Illustrations. Cloth, 21s. net. Second Impression.

"Every collector knows the name of Mr. Arthur Hayden, and knows him for a wise counsellor. Upon old furniture, old china, old pottery, and old prints there is no more knowing judge in the country; and in his latest volume he supplies a notable need, in the shape of a vade-mecum exploring some of the nondescript and little traversed bye-paths of the collector. There was never a time when the amateur of the antique stood more in need of a competent guide. . . . The man who wishes to avoid the pitfalls of the fraudulent will find much salutary advice in Mr. Hayden's gossip pages. There are chests, for example, a fruitful field for reproduction. Mr. Hayden gives photographs of many exquisite examples. There is a marriage coffer of the sixteenth century, decorated with carved figures of Cupid and Hymen, a fine Gothic chest of the fifteenth century, with rich foliated decorations; and a superb livery cupboard from Haddon Hall. From Flanders come steel coffers, with a lock of four bolts, the heavy sides strongly braized together. Then there are snuffers, with and without trays, tinder-boxes, snuff graters, and metal tobacco stoppers. The most fascinating designs are shown, with squirrels, dogs, and quaint human figures at the summit. Fans and playing-cards provide another attractive section.

Chicken-skin, delicate, white,

Painted by Carlo van Loo.

The fan has always been an object of the collector's passion, because of the grace of the article and its beauty as a display. Mr. Hayden shows a particularly beautiful one, with designs after Fragonard, the sticks of ivory with jewelled studs. Then there are watch-stands, a little baroque in design, and table-bells, some of them shaped as female figures with spreading skirts, old toys and picture-books, and, of course, cradles, of which every English farm-house once boasted its local variety. Altogether the book abounds in inviting

pictures and curious information, and is certain of a large, appreciative public.” *Daily Telegraph*.

TO BE PUBLISHED IN THE AUTUMN, 1920

The Fan Book: Including Special Chapters on European Fans of the Seventeenth and Eighteenth Centuries. By MACIVER PERCIVAL, author of “Chats on Old Jewellery and Trinkets.” Fully Illustrated. Demy 8vo, cloth, 21/- net.

POETRY THAT THRILLS

A COLLECTION OF SONGS FROM OVERSEAS THAT
THRILL WITH VIVID DESCRIPTIONS OF THE
ADVENTUROUS LIFE IN THE FROZEN NORTH, IN THE
OUTPOSTS OF CIVILIZATION AND OF THE HEROISM OF
SOLDIERS IN BATTLE

SONGS OF A SOURDOUGH. By ROBERT W. SERVICE. Crown 8vo. Cloth, 4/6 net. Thirty-ninth Impression. Also a Pocket edition. Fcap. 8vo, cloth, 4/6 net.

“Of the Canadian disciples of Kipling, by far the best is R. W. Service. His ‘Songs of a Sourdough’ have run through many editions. Much of his verse has a touch of real originality, conveying as it does a just impression of the something evil and askew in the strange, uncouth wilderness of the High North.” *The Times*.

“Mr. Service has got nearer to the heart of the old-time place miner than any other verse-maker in all the length and height of the Dominion. . . . He certainly sees the Northern Wilderness through the eyes of the man into whose soul it is entered.” *Morning Post*.

RHYMES OF A RED-CROSS MAN. By ROBERT W. SERVICE. Crown 8vo. Cloth, 4/6 net. Fifth Impression. Also a Pocket edition. Fcap. 8vo, cloth, 4/6 net.

“It is the great merit of Mr. Service’s verses that they are literally alive with the stress and joy and agony and hardship that make up life out in the battle zone. He has never written better than in this book, and that is saying a great deal.” *Bookman*.

“Mr. Service has painted for us the unutterable tragedy of the war, the horror, the waste, and the suffering, but side by side with that he has set the heroism, the endurance, the unfailing cheerfulness and the unquenchable laughter.” *Scots Pictorial*..

BALLADS OF A CHEECHAKO. By ROBERT W. SERVICE. Crown 8vo. Cloth, 4/6 net. Thirteenth Impression. Also a Pocket edition. Fcap. 8vo. Cloth, 4/6 net.

“It is to men like Mr. Service that we must look for really original verse nowadays; to the men on the frontiers of the world. ‘Ballads of a Cheechako’ is magnificent.” *Oxford Magazine*.

“All are interesting, arresting, and worth reading in their own setting for their own sakes. They are full of life and fire and muscularity, like the strenuous and devil-may-care fight of a life they describe.” *Standard*.

RHYMES OF A ROLLING STONE. By ROBERT W. SERVICE. Crown 8vo. Cloth, 4/6 net. Fourteenth Impression. Also a Pocket Edition. Fcap. 8vo. Cloth, 4/6 net.

“There is real rollicking fun in some of the rhymed stories, and some sound philosophy in the shorter serious poems which shows that Mr. Service is as many steps above the ordinary lesser poets in his thought as he is in his accomplishments.” *Academy*.

“Mr. Robert Service is, we suppose, one of the most popular verse-writers in the world. His swinging measures, his robust ballads of the outposts, his joy of living have fairly caught the ear of his countrymen.” *Spectator*.

THE SPELL OF THE TROPICS. By RANDOLPH H. ATKIN. Cloth, 4/6 net.

The poems are striking pen-pictures of life as it is lived by those men of the English-speaking races whose lot is cast in the sun-bathed countries of Latin-America. Mr. Atkin’s verses will reach the hearts of all who feel the call of the wanderlust, and, having shared their pleasures and hardships, his poems will vividly recall to “old-timers” bygone memories of days spent in the Land of the Coconut Tree.

THE SONG OF TIADATHA. By OWEN RUTTER. Cloth, 4/6 net. Third Impression.

Composed on the familiar metre of “Hiawatha,” “The Song of Tiadatha” (Tired Arthur), an extravaganza written in the highest spirits, nevertheless is

an epic of the war. It typifies what innumerable soldiers have seen and done and the manner in which they took it.

“This song of Tiadatha is nothing less than a little English epic of the war.” *The Morning Post*.

“Every Army officer and ex-officer will hail Tiadatha as a brother. ‘The Song of Tiadatha’ is one of the happiest skits born of the war.” *Evening Standard*.

SONGS OUT OF EXILE: Being Verses of African Sunshine and Shadow and Black Man’s Twilight. By CULLEN GOULDSBURY. Cloth, 4/6 net. Fourth Impression.

“The ‘Rhodesian Rhymes’ won for their author the journalistic title of ‘The Kipling of South Africa,’ and indeed his work is full of crisp vigour, fire and colour. It is brutal in parts; but its brutality is strong and realistic. Mr. Gouldsbury has spent many years in Rhodesia, and its life, black and white, is thoroughly familiar to him. . . . Mr. Gouldsbury is undoubtedly a writer to be reckoned with. His verse is informed by knowledge of wild life in open places and a measure of genuine feeling which make it real poetry.”—*Standard*.

FROM THE OUTPOSTS. By CULLEN GOULDSBURY. Cloth, 4/6 net. Third Impression.

“Mr. Cullen Gouldsbury’s collections of his verses are always welcome, and the last, ‘From the Outposts,’ is as good as its predecessor. No one has quite Mr. Gouldsbury’s experience and gift.” *Spectator*.

“It has been well said that Mr. Gouldsbury has done for the white man in Africa what Adam Lindsay Gordon in a measure accomplished for the Commonwealth and Kipling triumphantly for the British race, and he certainly is good to read.” *Field*.

THE HELL-GATE OF SOISSONS and other Poems. (“THE SONG OF THE GUNS.”) By HERBERT KAUFMAN. Cloth, 4/6 net. Fifth Impression.

“A singular gift for expressing in verse the facts, the heroism, even the humours of war; and in some cases voices its ideals with real eloquence.” *The Times*.

“Mr. Kaufman has undoubtedly given us a book worthy of the great hour that has brought it forth. He is a poet with a martial spirit and a deep, manly voice.” *Daily Mail*.

LYRA NIGERIA. By ADAMU. (E. C. ADAMS). Cloth, 4/6 net. Second Impression.

“Mr. E. C. Adams (Adamu) is a singer of Nigeria, and it can safely be said he has few, if any, rivals. There is something in these illustrations of Nigerian life akin to the style of Kipling and Service. The heart of the wanderer and adventurer is revealed, and in particular

that spirit of longing which comes to all . . . who have gone out to the far-lands of the world.” *Dundee Advertiser*.

SUNNY SONGS. Poems. By EDGAR A. GUEST. Cloth, 4/6 net.

In America Mr. Guest is an extraordinarily popular writer of verses, though this is his first introduction in book form to the British public. He brims over with sound sense and tonic cheeriness. He is keenly sensible of the humour of domestic life, but is deeply sympathetic with the associations which combine in the word “Home.” Hence he is read by women with amusement and pleasure. During the war his poem, “Said the Workman to the Soldier,” circulated by the hundred thousand. Like Béranger and all successful poets, he is essentially lyrical; that is to say, there is tune and swing in all his verses.

RICHARD MIDDLETON’S WORKS

POEMS AND SONGS (*First Series*). By RICHARD MIDDLETON. Cloth, 5/- net.

“We have no hesitation in placing the name of Richard Middleton beside the names of all that galaxy of poets that made the later Victorian era the most brilliant in poetry that England had known since the Elizabethan.” *Westminster Review*.

POEMS AND SONGS (*Second Series*). By RICHARD MIDDLETON. Cloth, 5/- net.

“Their beauty is undeniable and often of extraordinary delicacy, for Middleton had a mastery of craftsmanship such as is usually given to men of a far wider imaginative experience.” *Poetry Review*.

“Among the ‘Poems and Songs’ of Richard Middleton are to be found some of the finest of contemporary lyrics.” *Country Life*.

OTHER WORKS BY RICHARD MIDDLETON

THE GHOST SHIP AND OTHER STORIES.

MONOLOGUES.

THE DAY BEFORE YESTERDAY.

THE WAITING WOMAN and other Poems. By HERBERT KAUFMAN. Cloth, 4/6 net.

“Mr. Kaufman’s work possesses in a high degree the qualities of sincerity and truth, and it therefore never fails to move the reader. . . . This volume, in short, is the work of a genuine poet and artist.” *Aberdeen Free Press*.

“A versifier of great virility and power.” *Review of Reviews*.

BY W. B. YEATS AND OTHERS

POEMS. By W. B. YEATS. Second edition. Large Crown 8vo, Cloth, 10/6 net. Ninth Impression.

“Love songs, faery themes, moods of meditation, scenes of legendary wonder . . . is it possible that they should become so infinitely thrilling, touching, haunting in their fresh treatment, as though they had never been, or poets had never turned to them? In this poet’s hands they do so become. Mr. Yeats has given us a new thrill of delight, a new experience of beauty.” *Daily Chronicle*.

OTHER POEMS BY W. B. YEATS

COUNTESS CATHLEEN. A Dramatic Poem. Paper cover, 2/- net.

THE LAND OF HEART’S DESIRE. Paper cover, 1/6 net.

WHY DON’T THEY CHEER? By R. J. C. STEAD. Cloth, 4/6 net.

“Before the war Mr. Stead was known to Canadians as ‘The Poet of the Prairies.’ He must now be ranked as a ‘Poet of the Empire.’ . . . There is a strength, a beauty, a restrained passion in his war verses which prove his ability to penetrate into the heart of things such as very few of our war poets have exhibited.”—*Daily Express*.

SWORDS AND FLUTES. By WILLIAM KEAN SEYMOUR. Cloth, 4/- net.

“Among the younger poets Mr. Seymour is distinguished by his delicacy of technique. ‘Swords and Flutes’ is a book of grave and tender beauty expressed in lucent thought and jewelled words. ‘The Ambush’ is a lyric of mastery and fascination, alike in conception and rhythm, which should be included in any representative anthology of Georgian poetry.”
Daily Express.

THE MERMAID SERIES

THE BEST PLAYS OF THE OLD DRAMATISTS

Literal Reproductions of the Old Text. With Photogravure Frontispieces. Thin Paper edition. Boards, 3/-net; Cloth, 5/- net per volume. Leather, 7/6 net.

Marlowe. THE BEST PLAYS OF CHRISTOPHER MARLOWE. Edited, with Critical Memoir and Notes, by Havelock Ellis; and containing a General Introduction to the Series by John Addington Symonds.

Otway. THE BEST PLAYS OF THOMAS OTWAY. Introduction and Notes by the Hon. Roden Noel.

Ford. THE BEST PLAYS OF JOHN FORD. Edited by Havelock Ellis.

Massinger. THE BEST PLAYS OF PHILLIP MASSINGER. With Critical and Biographical Essay and Notes by Arthur Symonds.

Heywood (T.). THE BEST PLAYS OF THOMAS HEYWOOD. Edited by A. W. Verity. With Introduction by J. A. Symonds.

Wycherley. THE COMPLETE PLAYS OF WILLIAM WYCHERLEY. Edited, with Introduction and Notes, by W. C. Ward.

NERO AND OTHER PLAYS. Edited by H. P. Horne, Arthur Symonds, A. W. Verity and H. Ellis.

Beaumont. THE BEST PLAYS OF BEAUMONT AND FLETCHER. Introduction and Notes by J. St. Loe Strachey. 2 vols.

Congreve. THE COMPLETE PLAYS OF WILLIAM CONGREVE. Edited by Alex. C. Ewald.

Symonds (J. A.). THE BEST PLAYS OF WEBSTER AND TOURNEUR. With an Introduction and Notes by John Addington Symonds.

Middleton (T.). THE BEST PLAYS OF THOMAS MIDDLETON. With an Introduction by Algernon Charles Swinburne. 2 vols.

Shirley. THE BEST PLAYS OF JAMES SHIRLEY. With Introduction by Edmund Gosse.

Dekker. THE BEST PLAYS OF THOMAS DEKKER. Notes by Ernest Rhys.

Steele (R.). THE COMPLETE PLAYS OF RICHARD STEELE. Edited, with Introduction and Notes, by G. A. Aitken.

Jonson. THE BEST PLAYS OF BEN JONSON. Edited, with Introduction and Notes, by Brinsley Nicholson and C. H. Herford. 2 vols.

Chapman. THE BEST PLAYS OF GEORGE CHAPMAN. Edited by William Lyon Phelps.

Vanbrugh. THE SELECT PLAYS OF SIR JOHN VANBRUGH. Edited, with an Introduction and Notes, by A. E. H. Swain.

Shadwell. THE BEST PLAYS OF THOMAS SHADWELL. Edited by George Saintsbury.

Dryden. THE BEST PLAYS OF JOHN DRYDEN. Edited by George Saintsbury. 2 vols.

Farquhar. THE BEST PLAYS OF GEORGE FARQUHAR. Edited, and with an Introduction, by William Archer.

Greene. THE COMPLETE PLAYS OF ROBERT GREENE. Edited with
Introduction and Notes by Thomas H. Dickinson.

THE ADVANCE OF SOUTH AMERICA
A FEW NOTES ON SOME INTERESTING BOOKS DEALING WITH
THE PAST HISTORY, PRESENT AND FUTURE POSSIBILITIES OF
THE GREAT CONTINENT

When in 1906 Mr. Fisher Unwin commissioned the late Major Martin Hume to prepare a series of volumes by experts on the South American Republics, but little interest had been taken in the country as a possible field for commercial development. The chief reasons for this were ignorance as to the trade conditions and the varied resources of the country, and the general unrest and instability of most of the governments. With the coming of the South American Series of handbooks the financial world began to realize the importance of the country, and, with more settled conditions, began in earnest to develop the remarkable natural resources which awaited outside enterprise. Undoubtedly the most informative books on the various Republics are those included in THE SOUTH AMERICAN SERIES, each of which is the work of a recognized authority on his subject.

“The output of books upon Latin America has in recent years been very large, a proof doubtless of the increasing interest that is felt in the subject. Of these the ‘South American Series’ is the most noteworthy.” *The Times*.

“When the ‘South American Series’ is completed, those who take interest in Latin-American affairs will have an invaluable encyclopædia at their disposal.” *Westminster Gazette*.

“Mr. Unwin’s ‘South American Series’ of books are of special interest and value to the capitalist and trader.”—*Chamber of Commerce Journal*.

Full particulars of the volumes in the “South American Series,” also of other interesting books on South America, will be found in the pages following.

1 **Chile.** By G. F. SCOTT ELLIOTT, M.A., F.R.G.S. With an Introduction by MARTIN HUME, a Map and 89 Illustrations. Cloth, 15/- net. Fifth Impression.

“An exhaustive, interesting account, not only of the turbulent history of this country, but of the present conditions and seeming prospects.” *Westminster Gazette*.

2 **Peru.** By C. REGINALD ENOCK, F.R.G.S. With an Introduction by MARTIN HUME, a Map and 64 Illustrations. Cloth, 18/- net. Fifth Impression.

“An important work. . . . The writer possesses a quick eye and a keen intelligence; is many-sided in his interests, and on certain subjects speaks as an expert. The volume deals fully with the development of the country.” *The Times*.

3 **Mexico.** By C. REGINALD ENOCK, F.R.G.S. With an Introduction by MARTIN HUME, a Map and 64 Illustrations. Cloth, 15/- net. Fifth Impression.

“The book is most comprehensive; the history, politics, topography, industries, resources and possibilities being most ably discussed.” *The Financial News*.

4 **Argentina.** By W. A. HIRST. With an Introduction by Martin Hume, a Map and 64 Illustrations. Cloth, 15/- net. Fifth Impression.

“The best and most comprehensive of recent works on the greatest and most progressive of the Republics of South America.” *Manchester Guardian*.

5 **Brazil.** By PIERRE DENIS. Translated, and with an Historical Chapter by BERNARD MIALL. With a Supplementary Chapter by DAWSON A. VINDIN, a Map and 86 Illustrations. Cloth, 15/- net. Third Impression.

“Altogether the book is full of information, which shows the author to have made a most careful study of the country.”—*Westminster Gazette*.

6 **Uruguay.** By W. H. KOEBEL. With a Map and 55 Illustrations. Cloth, 15/- net. Third Impression.

“Mr. Koebel has given us an expert’s diagnosis of the present condition of Uruguay. Glossing over nothing, exaggerating nothing, he has prepared a document of the deepest interest.”—*Evening Standard*.

7 **Guiana.** British, French and Dutch. By JAMES RODWAY. With a Map and 82 Illustrations. Cloth, 18/- net. Second Impression.

“Mr. Rodway’s work is a storehouse of information, historical, economical and sociological.” *The Times*.

8 **Venezuela.** By LEONARD V. DALTON, F.G.S., F.R.G.S. With a Map and 45 Illustrations. Cloth, 15/- net. Third Impression.

“An exhaustive and valuable survey of its geography, geology, history, botany, zoology and anthropology, and of its commercial possibilities in the near future.” *Manchester Guardian*.

9 **Latin America:** Its Rise and Progress. By F. GARCIA-CALDERON. With a Preface by RAYMOND POINCARÉ, President of the French Republic. With a Map and 84 Illustrations. Cloth, 15/- net. Fifth Impression.

President Poincaré, in a striking preface to this book, says: “Here is a book that should be read and digested by every one interested in the future of the Latin genius.”

10 **Colombia.** By PHANOR JAMES EDER, A.B., LL.B. With 2 Maps and 40 Illustrations. Cloth, 15/- net. Fourth Impression.

“Mr. Eder’s valuable work should do much to encourage investment, travel and trade in one of the least-known and most promising of the countries of the New World.” *Manchester Guardian*.

11 **Ecuador.** By C. REGINALD ENOCK, F.R.G.S. With 2 Maps and 87 Illustrations. Cloth, 15/- net. Second Impression.

“Mr. Enock’s very thorough and exhaustive volume should help British investors to take their part in promoting its development. He has studied and described the country in all its aspects.” *Manchester Guardian*.

12 **Bolivia.** By PAUL WALLE. With 4 Maps and 59 Illustrations. Cloth, 18/- net. Second Impression.

Bolivia is a veritable El Dorado, requiring only capital and enterprise to become one of the wealthiest States of America. This volume is the result of a careful investigation made on behalf of the French Ministry of Commerce.

13 **Paraguay.** By W. H. KOEBEL. With a Map and 32 Illustrations. Cloth, 15/- net. Second Impression.

“Gives a great deal of serious and useful information about the possibilities of the country for the emigrant, the investor and the tourist, concurrently with a vivid and literary account of its history.” *Economist*.

14 Central America: Guatemala, Nicaragua, Costa Rica, Honduras, Panama and Salvador. By W. H. KOEBEL. With a Map and 25 Illustrations. Cloth, 15/- net. Second Impression.

“We strongly recommend this volume, not only to merchants looking ahead for new openings for trade, but also to all who wish for an accurate and interesting account of an almost unknown world.” *Saturday Review*.

BOOKS ON SOUTH AMERICA

OTHER BOOKS ON SOUTH AMERICA

Spanish America: Its Romance, Reality and Future. By C. R. ENOCK, Author of “The Andes and the Amazon,” “Peru,” “Mexico,” “Ecuador.” Illustrated and with a Map. 2 vols. Cloth, 30/- net the set.

Starting with the various States of Central America, Mr. Enock then describes ancient and modern Mexico, then takes the reader successively along the Pacific Coast, the Cordillera of the Andes, enters the land of the Spanish Main, conducts the reader along the Amazon Valley, gives a special chapter to Brazil and another to the River Plate and Pampas. Thus all the States of Central and South America are covered. The work is topographical, descriptive and historical; it describes the people and the cities, the flora and fauna, the varied resources of South America, its trade, railways, its characteristics generally.

South America: An Industrial and Commercial Field. By W. H. KOEBEL. Illustrated. Cloth, 18/- net. Second Impression.

“The book considers such questions as South American commerce, British interests in the various Republics, international relations and trade, communications, the tendency of enterprise, industries, etc. Two chapters devoted to the needs of the continent will be of especial interest to manufacturers and merchants, giving as they do valuable hints as to the various goods required, while the chapter on merchandise and commercial travellers affords some sound and practical advice.” *Chamber of Commerce Journal*.

Vagabonding down the Andes. By HARRY A. FRANCK, author of “A Vagabond Journey Round the World,” etc. With a Map and 176 Illustrations. Cloth, 25/- net. Second Impression.

“The book is a brilliant record of adventurous travel among strange scenes and with even more strange companions, and vividly illustrates, by its graphic text and its admirable photographs, the real conditions of life in the backwood regions of South America.”

Manchester Guardian.

“Mr. Franck is to be congratulated on having produced a readable and even fascinating book. His journey lay over countries in which an increasing interest is being felt. Practically speaking, he may be said to have started from Panama, wandered through Colombia, spending some time at Bogota, and then going on to Ecuador, of which Quito is the centre. Next he traversed the fascinating country of the Incas, from the borders of which he entered Bolivia, going right across that country till he approached Brazil. He passed through Paraguay, cut through a corner of the Argentine to Uruguay, and so to the River Plata and the now well-known town of Buenos Ayres.” *Country Life.*

In the Wilds of South America: Six Years of Exploration in Colombia, Venezuela, British Guiana, Peru, Bolivia, Argentina, Paraguay and Brazil. By LEO E. MILLER, of the American Museum of Natural History. With 48 Full-page Illustrations and with Maps. Cloth, 21/- net.

This volume represents a series of almost continuous explorations hardly ever paralleled in the huge areas traversed. The author is a distinguished field naturalist—one of those who accompanied Colonel Roosevelt on his famous South American expedition—and his first object in his wanderings over 150,000 miles of territory was the observation of wild life; but hardly second was that of exploration. The result is a wonderfully informative, impressive and often thrilling narrative in which savage peoples and all but unknown animals largely figure, which forms an infinitely readable book and one of rare value for geographers, naturalists and other scientific men.

The Putumayo: The Devil's Paradise. Travels in the Peruvian Amazon Region and an Account of the Atrocities committed upon the Indians therein. By E. W. HARDENBURG, C.E. Edited and with an Introduction by C. REGINALD ENOCK, F.R.G.S. With a Map and 16 Illustrations. Demy 8vo, Cloth, 10/6 net. Second Impression.

“The author gives us one of the most terrible pages in the history of trade.” *Daily Chronicle.*

Tramping through Mexico, Guatemala and Honduras. By HARRY A. FRANCK, With a Map and 88 Illustrations. Cloth, 7/6 net.

“Mr. Harry Franck is a renowned vagabond with a gift for vivid description. . . . His record is well illustrated and he tells his story in an attractive manner, his descriptions of scenery being so well done that one feels almost inclined to risk one’s life in a wild race dwelling in a land of lurid beauty.” *Liverpool Mercury*.

“Mr. Franck has combined with an enthralling and amusing personal narrative a very vivid and searching picture, topographical and social, of a region of much political and economic interest.” *Glasgow Herald*.

Mexico (STORY OF THE NATIONS). By SUSAN HALE. With Maps and 47 Illus. Cloth, 7/6 net. Third Impression.

“This is an attractive book. There is a fascination about Mexico which is all but irresistible. . . . The authoress writes with considerable descriptive power, and all through the stirring narrative never permits us to lose sight of natural surroundings.” *Dublin Review*.

Things as they are in Panama. By HARRY A. FRANCK. With 50 Illustrations. Cloth, 7/6 net.

“Mr. Franck writes from personal knowledge, fortified by the aptitude of a practical and shrewd observer with a sense of humour, and the result is a word-picture of unusual vividness.”—*Standard*.

“A sparkling narrative which leaves one wondering again why the general reader favours modern fiction so much when it is possible to get such vivacious yarns as this about strange men and their ways in a romantic corner of the tropics.” *Daily Mail*.

The Spell of the Tropics. POEMS. By RANDOLPH H. ATKIN. Cloth, 4/6 net.

The author has travelled extensively in Central and South America, and has strongly felt the spell of those tropic lands, with all their splendour and romance, and yet about which so little is known. The poems are striking pen-pictures of life as it is lived by those men of the English-speaking races whose lot is cast in the sun-bathed countries of Latin-America. Mr. Atkin’s verses will reach the hearts of all who feel the call of the wanderlust, and, having shared their pleasures and hardships, his poems will vividly recall to “old-timers” bygone memories of days spent in the land of the Coconut Tree.

Baedeker Guide to the United States. With Excursions to Mexico, Cuba, Porto Rico and Alaska. With 33 Maps and 48 Plans. Fourth Edition, 1909. Cloth, 20/- net.

IMPORTANT. Travellers to the Republics of South America will find WESSELY'S ENGLISH-SPANISH and SPANISH-ENGLISH DICTIONARY and WESSELY'S LATIN-ENGLISH and ENGLISH-LATIN DICTIONARY invaluable books. Bound in cloth, pocket size. Price 4/- net each.

Ask for Wessely's Edition, published by Mr. T. Fisher Unwin.

THE STORY OF THE NATIONS

THE GREATEST HISTORICAL LIBRARY IN THE WORLD :: :: 67
VOLUMES

Each volume of "The Story of the Nations" Series is the work of a recognized scholar, chosen for his knowledge of the subject and ability to present history in an attractive form, for the student and the general reader. The Illustrations and Maps are an attractive feature of the volume, which are strongly bound for constant use.

67 Volumes. Cloth, 7s. 6d. net each.

See page 2 for an announcement of a new volume.

"It is many years since Messrs. T. Fisher Unwin commenced the publication of a series of volumes now entitled "The Story of the Nations." Each volume is written by an acknowledged authority on the country with which it deals. The series has enjoyed great popularity, and not an uncommon experience being the necessity for a second, third, and even fourth impression of particular volumes." *Scotsman*.

"Probably no publisher has issued a more informative and valuable series of works than those included in "The Story of the Nations." " *To-Day*.

"The series is likely to be found indispensable in every school library." *Pall Mall Gazette*.

"An admirable series." *Spectator*.

“Such a universal history as the series will present us with in its completion will be a possession such as no country but our own can boast of. Its success on the whole has been very remarkable.” *Daily Chronicle*.

“There is perhaps no surer sign of the increased interest that is now being taken in historical matters than the favourable reception which we believe both here and in America is being accorded to the various volumes of “The Story of the Nations” as they issue in quick succession from the press. More than one volume has reached its third edition in England alone. . . . Each volume is written by one of the foremost English authorities on the subject with which it deals. . . . It is almost impossible to over-estimate the value of the series of carefully prepared volumes, such as are the majority of those comprising this library. . . . The illustrations make one of the most attractive features of the series.”—*Guardian*.

A NEW VOLUME IN “THE STORY OF THE NATIONS” READY IN THE AUTUMN, 1920

BELGIUM FROM THE ROMAN INVASION TO THE PRESENT DAY

By EMILE CAMMAERTS. With Maps and Illustrations. Large Crown 8vo. Cloth, 7/6 net.

A complete history of the Belgian nation from its origins to its present situation has not yet been published in this country. Up till now Belgian history has only been treated as a side issue in works concerned with the Belgian art, Belgian literature or social conditions. Besides, there has been some doubt with regard to the date at which such a history ought to begin, and a good many writers have limited themselves to the modern history of Belgium because they did not see in olden times sufficient evidence of Belgian unity. According to the modern school of Belgian historians, however, this unity, founded on common traditions and common interests, has asserted itself again and again through the various periods of history in spite of invasion, foreign domination and the various trials experienced by the country. The history of the Belgian nation appears to the modern mind as a slow development of one nationality constituted by two races speaking two different languages but bound together by geographical, economic and cultural conditions. In view of the recent proof Belgium has given of her patriotism during the world-war, this impartial enquiry into her origins may

prove interesting to British readers. Every opportunity has been taken to insist on the frequent relationships between the Belgian provinces and Great Britain from the early middle ages to the present time, and to show the way in which both countries were affected by them. Written by one of the most distinguished Belgian writers, who has made a specialty of his subject, this work will be one of the most brilliant and informing contributions in “The Story of the Nations.”

A COMPLETE LIST OF THE VOLUMES IN “THE STORY OF
THE NATIONS” SERIES. THE FIRST AND MOST COMPLETE
LIBRARY OF THE WORLD’S HISTORY PRESENTED IN A
POPULAR FORM

- 1 **Rome:** From the Earliest Times to the End of the Republic. By ARTHUR GILMAN, M.A. Third Edition. With 48 Illustrations and Maps.
- 2 **The Jews:** In Ancient, Mediæval and Modern Times. By Professor JAMES K. HOSMER. Eighth Impression. With 37 Illustrations and Maps.
- 3 **Germany.** By S. BARING-GOULD, M.A. Seventh Impression. With 108 Illustrations and Maps.
- 4 **Carthage: or the Empire of Africa.** By Professor ALFRED J. CHURCH, M.A. With the Collaboration of Arthur Gilman, M.A. Ninth Impression. With 43 Illustrations and Maps.
- 5 **Alexander’s Empire.** By JOHN PENTLAND MAHAFFY, D.D. With the Collaboration of Arthur Gilman, M.A. Eighth Impression. With 43 Illustrations and Maps.
- 6 **The Moors in Spain.** By STANLEY LANE-POOLE. With the Collaboration of Arthur Gilman, M.A. Eighth Edition. With 29 Illustrations and Maps.
- 7 **Ancient Egypt.** By Professor GEORGE RAWLINSON, M.A. Tenth Edition. Eleventh Impression. With 50 Illustrations and Maps.

- 8 **Hungary.** In Ancient, Mediæval and Modern Times. By Professor ARMINIUS VAMBÉRY. With Collaboration of Louis Heilpin. Seventh Edition. With 47 Illustrations and Maps.
- 9 **The Saracens:** From the Earliest Times to the Fall of Bagdad. By ARTHUR GILMAN, M.A. Fourth Edition. With 57 Illustrations and Maps.
- 10 **Ireland.** By the Hon. EMILY LAWLESS. Revised and brought up to date by J. O'Toole. With some additions by Mrs. Arthur Bronson. Eighth Impression. With 58 Illustrations and Maps.
- 11 **Chaldea:** From the Earliest Times to the Rise of Assyria. By ZÉNAÏDE A. RAGOZIN. Seventh Impression. With 80 Illustrations and Maps.
- 12 **The Goths:** From the Earliest Times to the End of the Gothic Dominion in Spain. By HENRY BRADLEY. Fifth Edition. With 85 Illustrations and Maps.
- 13 **Assyria:** From the Rise of the Empire to the Fall of Nineveh. (Continued from "Chaldea.") By ZÉNAÏDE A. RAGOZIN. Seventh Impression. With 81 Illustrations and Maps.
- 14 **Turkey.** By STANLEY LANE-POOLE, assisted by C. J. W. Gibb and Arthur Gilman. New Edition. With a new Chapter on recent events (1908). With 43 Illustrations and Maps.
- 15 **Holland.** By Professor J. E. THOROLD ROGERS. Fifth Edition. With 57 Illustrations and Maps.
- 16 **Mediæval France:** From the Reign of Huguar Capet to the beginning of the 16th Century. By GUSTAVE MASSON, B.A. Sixth Edition. With 48 Illustrations and Maps.
- 17 **Persia.** By S. G. W. BENJAMIN. Fourth Edition. With 56 Illustrations and Maps.
- 18 **Phœnicia.** By Professor GEORGE RAWLINSON, M.A. Third Edition. With 47 Illustrations and Maps.

- 19 **Media, Babylon, and Persia:** From the Fall of Nineveh to the Persian War. By ZÉNAÏDE A. RAGOZIN. Fourth Edition. With 17 Illustrations and Maps.
- 20 **The Hansa Towns.** By HELEN ZIMMERN. Third Edition. With 51 Illustrations and Maps.
- 21 **Early Britain.** By Professor ALFRED J. CHURCH, M.A. Sixth Impression. With 57 Illustrations and Maps.
- 22 **The Barbary Corsairs.** By STANLEY LANE-POOLE. With additions by J. D. KELLY. Fourth Edition. With 39 Illustrations and Maps.
- 23 **Russia.** By W. R. MORFILL, M.A. Fourth Edition. With 60 Illustrations and Maps.
- 24 **The Jews under Roman Rule.** By W. D. MORRISON. Second Impression. With 61 Illustrations and Maps.
- 25 **Scotland:** From the Earliest Times to the Present Day. By JOHN MACKINTOSH, LL.D. Fifth Impression. With 60 Illustrations and Maps.
- 26 **Switzerland.** By LINA HUG and R. STEAD. Third Impression. With over 54 Illustrations, Maps, etc.
- 27 **Mexico.** By SUSAN HALE. Third Impression. With 47 Illustrations and Maps.
- 28 **Portugal.** By H. MORSE STEPHENS, M.A. New Edition. With a new Chapter by Major M. HUME and 5 new Illustrations. Third Impression. With 44 Illustrations and Maps.
- 29 **The Normans.** Told chiefly in Relation to their Conquest of England. By SARAH ORNE JEWETT. Third Impression. With 35 Illustrations and Maps.
- 30 **The Byzantine Empire.** By C. W. C. OMAN, M.A. Third Edition. With 44 Illustrations and Maps.
- 31 **Sicily:** Phœnician, Greek, and Roman. By Professor E. A. FREEMAN. Third Edition. With 45 Illustrations.

- 32 **The Tuscan Republics** (Florence, Siena, Pisa, Lucca) **with Genoa.** By BELLA DUFFY. With 40 Illustrations and Maps.
- 33 **Poland.** By W. R. MORFILL. Third Impression. With 50 Illustrations and Maps.
- 34 **Parthia.** By Professor GEORGE RAWLINSON. Third Impression. With 48 Illustrations and Maps.
- 35 **The Australian Commonwealth.** (New South Wales, Tasmania, Western Australia, South Australia, Victoria, Queensland, New Zealand.) By GREVILLE TREGARTHEN. Fifth Impression. With 36 Illustrations and Maps.
- 36 **Spain.** Being a Summary of Spanish History from the Moorish Conquest to the Fall of Granada (A.D. 711–1492). By HENRY EDWARD WATTS. Third Edition. With 36 Illustrations and Maps.
- 37 **Japan.** By DAVID MURRAY, Ph.D., LL.D. With a new Chapter by JOSEPH W. LONGFORD. 35 Illustrations and Maps.
- 38 **South Africa.** (The Cape Colony, Natal, Orange Free State, South African Republic, Rhodesia, and all other Territories south of the Zambesi.) By Dr. GEORGE MCCALL THEAL, D.Litt., LL.D. Revised and brought up to date. Eleventh Impression. With 39 Illustrations and Maps.
- 39 **Venice.** By ALETHEA WIEL. Fifth Impression. With 61 Illustrations and a Map.
- 40 **The Crusades:** The Story of the Latin Kingdom of Jerusalem. By T. A. ARCHER and C. L. KINGSFORD. Third Impression. With 58 Illustrations and 3 Maps.
- 41 **Vedic India:** As embodied principally in the Rig-Veda. By ZÉNAÏDE A. RAGOZIN. Third Edition. With 36 Illustrations and Maps.
- 42 **The West Indies and the Spanish Main.** By JAMES RODWAY, F.L.S. Third Impression. With 48 Illustrations and Maps.

- 43 **Bohemia:** From the Earliest Times to the Fall of National Independence in 1620; with a Short Summary of later Events. By C. EDMUND MAURICE. Second Impression. With 41 Illustrations and Maps.
- 44 **The Balkans** (Rumania, Bulgaria, Servia and Montenegro). By W. MILLER, M.A. New Edition. With a new Chapter containing their History from 1296 to 1908. With 39 Illustrations and Maps.
- 45 **Canada.** By Sir JOHN BOURINOT, C.M.G. With 63 Illustrations and Maps. Second Edition. With a new Map and revisions, and a supplementary Chapter by EDWARD PORRITT. Third Impression.
- 46 **British India.** By R. W. FRAZER, LL.D. Eighth Impression. With 30 Illustrations and Maps.
- 47 **Modern France, 1789–1895.** By ANDRÉ LEBON. With 26 Illustrations and a Chronological Chart of the Literary, Artistic, and Scientific Movement in Contemporary France. Fourth Impression.
- 48 **The Franks.** From their Origin as a Confederacy to the Establishment of the Kingdom of France and the German Empire. By LEWIS SERGEANT. Second Edition. With 40 Illustrations and Maps.
- 49 **Austria.** By SIDNEY WHITMAN. With the Collaboration of J. R. MCILRAITH. Third Edition. With 35 Illustrations and a Map.
- 50 **Modern England before the Reform Bill.** By JUSTIN MCCARTHY. With 31 Illustrations.
- 51 **China.** By Professor R. K. DOUGLAS. Fourth Edition. With a new Preface. 51 Illustrations and a Map. Revised and brought up to date by IAN C. HANNAH.
- 52 **Modern England under Queen Victoria:** From the Reform Bill to the Present Time. By JUSTIN MCCARTHY. Second Edition. With 46 Illustrations.
- 53 **Modern Spain, 1878–1898.** By MARTIN A. S. HUME. Second Impression. With 37 Illustrations and a Map.

- 54 **Modern Italy, 1748–1898.** By Professor PIETRO ORSI. With over 40 Illustrations and Maps.
- 55 **Norway:** From the Earliest Times. By Professor HJALMAR H. BOYESEN. With a Chapter by C. F. KEARY. With 77 Illustrations and Maps.
- 56 **Wales.** By OWEN EDWARDS. With 47 Illustrations and 7 Maps. Fifth Impression.
- 57 **Mediæval Rome:** From Hildebrand to Clement VIII, 1073–1535. By WILLIAM MILLER. With 35 Illustrations.
- 58 **The Papal Monarchy:** From Gregory the Great to Boniface VIII. By WILLIAM BARRY, D.D. Second Impression. With 61 Illustrations and Maps.
- 59 **Mediæval India under Mohammedan Rule.** By STANLEY LANE-POOLE. With 59 Illustrations. Twelfth Impression.
- 60 **Parliamentary England:** The Evolution of the Cabinet System, 1660–1832. By EDWARD JENKS. With 47 Illustrations.
- 61 **Buddhist India.** By T. W. RHYS DAVIDS. Fourth Impression. With 57 Illustrations and Maps.
- 62 **Mediæval England, 1066–1350.** By MARY BATESON. With 93 Illustrations.
- 63 **The Coming of Parliament.** (England, 1350–1660.) By L. CECIL JANE. With 51 Illustrations and a Map.
- 64 **The Story of Greece:** From the Earliest Times to A.D. 14. By E. S. SHUCKBURGH. With 2 Maps and about 70 Illustrations.
- 65 **The Story of the Roman Empire.** (29 B.C. to A.D. 476.) By H. STUART JONES. Third Impression. With a Map and 52 Illustrations.
- 66 **Sweden and Denmark.** With Chapters on Finland and Iceland. By JON STEFANSSON. With Maps and 40 Illustrations.
- 67 **Belgium.** (*See page 2 for special announcement.*)

IMPORTANT—ASK YOUR BOOKSELLER TO LET
YOU EXAMINE A SPECIMEN VOLUME OF
“THE STORY OF THE NATIONS” SERIES

T. FISHER UNWIN Ltd., 1 Adelphi Terrace, London, W.C.2
And of all Booksellers throughout the World

*** END OF THE PROJECT GUTENBERG EBOOK WHY DO WE DIE?

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.